Characterizing Heavy Neutral Leptons: Measuring Parameters, Discriminating Majorana versus Dirac, and Using FASER2 as a Trigger for ATLAS

Daniel La Rocco

[2510.16107](https://arxiv.org/abs/2510.16107)

Collaboration: Jonathan L. Feng, Alec Hewitt, ***Daniel La Rocco,*** *and Daniel Whiteson*

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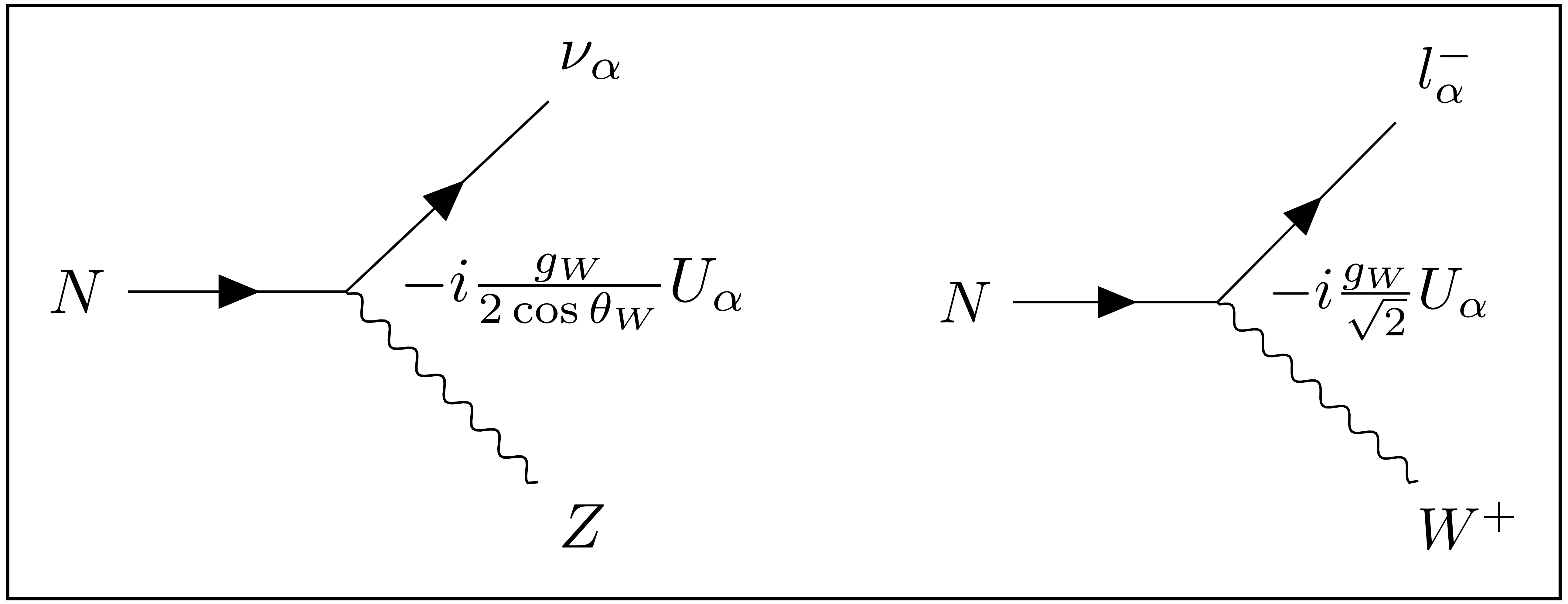
Heavy Neutral Leptons (HNLs)

- A simple solution to the neutrino mass problem is the introduction of a **right-handed sterile neutrino** (HNL if referring to the mass eigenstate)
- This allows for the inclusion of a **Dirac mass** through a Yukawa interaction after electroweak symmetry breaking (EWSB)
- Additionally, one can incorporate **Majorana mass** terms which violate lepton number
- After diagonalization, the mass eigenstates pick up suppressed neutral current (NC) and charged-current (CC) interactions with the Z and W bosons, respectively

$$\mathscr{L} \supset -\sum_{\alpha} \underbrace{Y_\alpha \bar{L}_\alpha \tilde{\phi} N'}_{\text{Dirac}} - \underbrace{M_R \overline{N'^c} N' - \sum_{\alpha\beta} \frac{C_5^{\alpha\beta}}{\Lambda} \bar{L}_\alpha \tilde{\phi} \tilde{\phi}^T L_\beta^c}_{\text{Majorana}} + \text{h.c.}$$

$$M_\nu = \begin{pmatrix} M_L & M_D \\ M_D^T & M_R \end{pmatrix} \rightarrow \nu_\alpha = \sum_{i=1}^{3} V_{\alpha i} \nu_i + U_\alpha N^c$$



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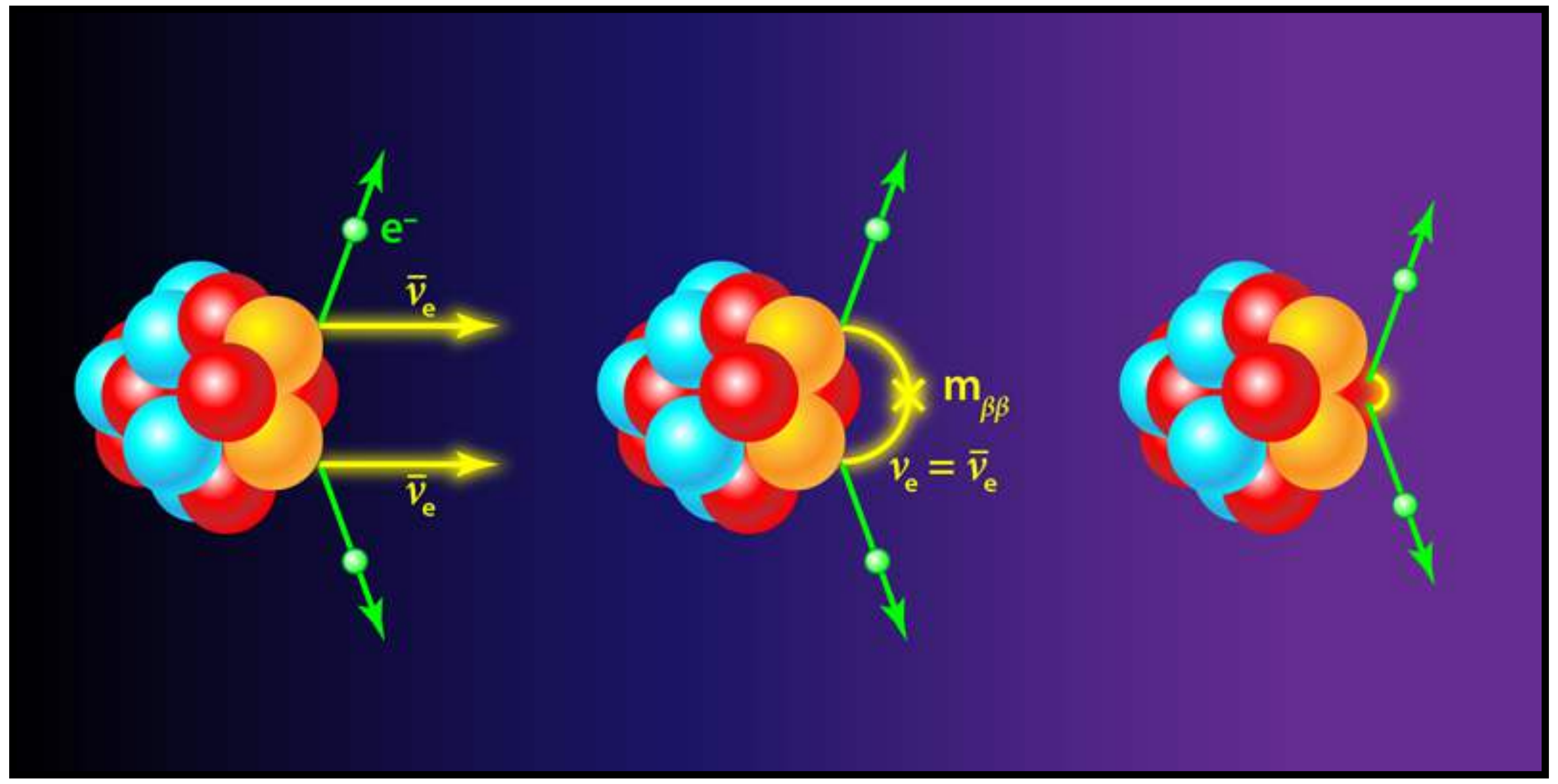
Majorana or Dirac?

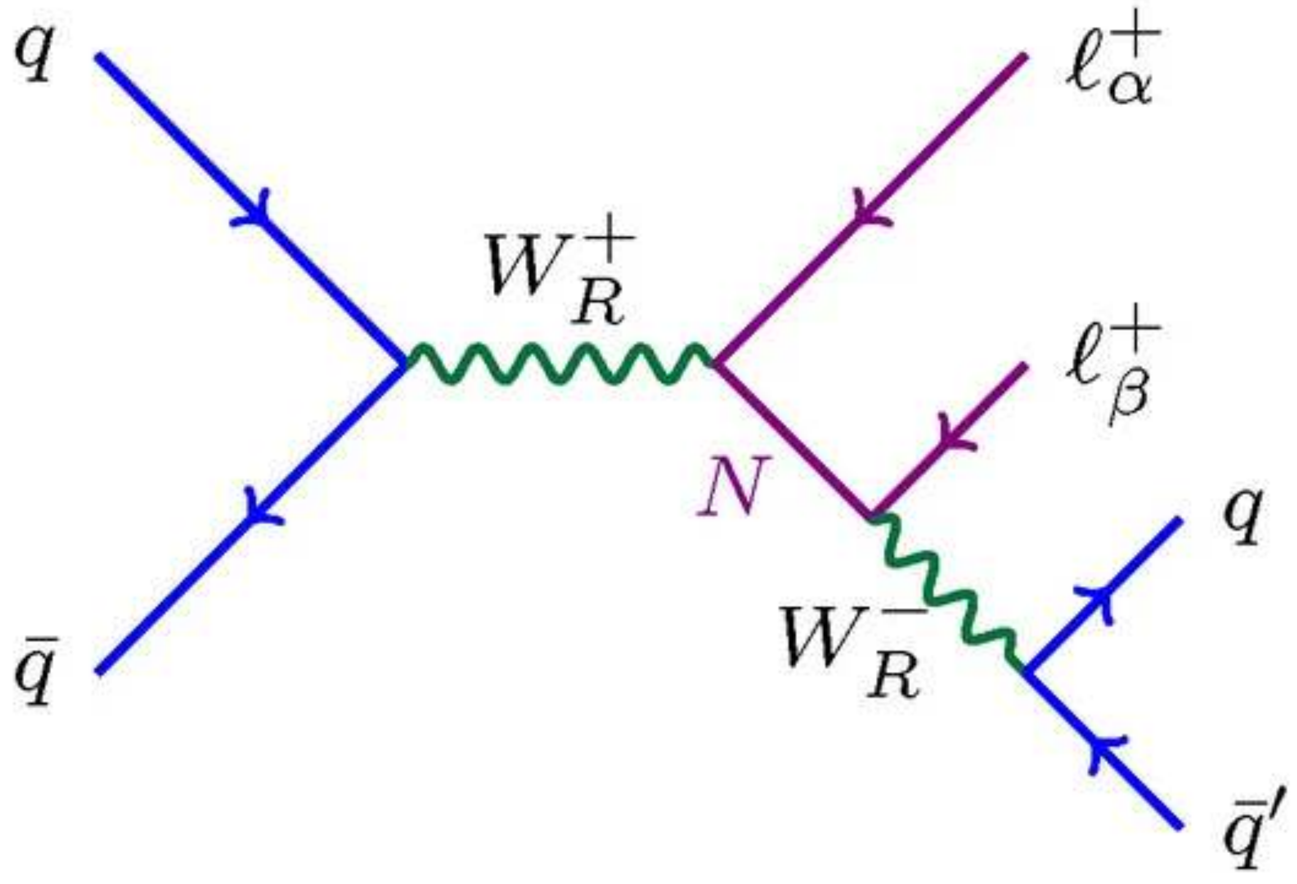
- Some important questions:
  - are neutrinos Dirac or Majorana?
  - is lepton number violation (LNV) allowed in the SM?
    - Implications for cosmology, neutrino oscillations, and dark matter
- Direct tests of LNV have been carried out for decades in lab and collider based experiments:
  - $0\nu\beta\beta$ (Furry 1939)
  - $pp \to l^{\pm}l^{\pm}jj$ (Keung and Senjanović 1983)

Neutrinoless double beta decay



[https://physics.aps.org/articles/v11/30]

$pp \to l^{\pm}l^{\pm}jj$



[https://www.mdpi.com/2218-1997/8/3/164]

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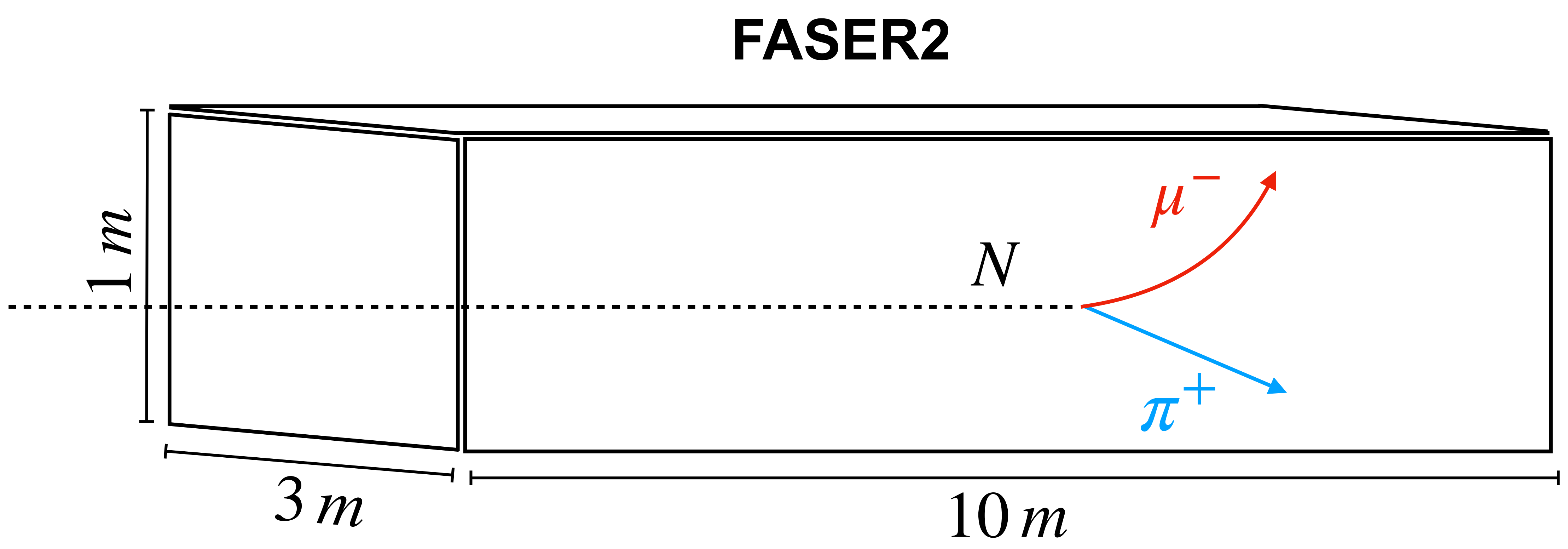
Characterizing New Physics

- In the event an HNL is detected, much work will be done to identify the underlying nature of the observed signal excess.
- In our work, **we assume that a discovery has been made at FASER2** and answer the following questions:
  - How well can FASER2 estimate the HNL model parameters?
  - Is it possible to discriminate between Dirac and Majorana HNLs using event kinematics?
  - Can FASER2 be used as a trigger for ATLAS to measure lepton number violation?
- Using the MC event generator FORESEE, we perform an asymptotic likelihood-based analysis on MC-generated HNL data



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Benchmark Models

- Simplifying Assumptions:
  - $U_e = U_\tau = 0$
  - $N \to \mu\pi$ only
- **Model 1 (High Statistics):**
  - 1.84 GeV Majorana HNL
  - $N \sim 8000$
- **Model 2 (Low Statistics):**
  - 2.0 GeV Majorana HNL
  - $N \sim 80$

		Model 1		Model 2		
HNL Model Parameters		$m_N = 1.84$ GeV, $U_\mu = 0.0036$, Majorana		$m_N = 2.00$ GeV, $U_\mu = 0.002$, Majorana		
$c\tau_N$		2.03 m		4.33 m		
$\langle E_N \rangle_{\text{FASER2}}$		820 GeV		828 GeV		
$\langle E_N \rangle_{\text{FASER2+ATLAS}}$		458 GeV		452 GeV		
			FASER2		FASER2	FASER2+ATLAS
Production Modes	N_{events} at HL-LHC (in 3 ab^{-1})	$D_s \to \mu N$	119,000	$B \to D^{*0}\mu N$	447	56.1
		$B \to D^{*0}\mu N$	1670	$B^0 \to D^{*}\mu N$	405	50.7
		$B^0 \to D^{*}\mu N$	1570	$B \to D^0\mu N$	199	26.4
		$B_s^0 \to D_s^{*}\mu N$	333	$B^0 \to D\mu N$	183	22.4
Decay Modes	Branching Fraction	$\mu\rho$	0.187	$\mu\rho$	0.162	
		$\nu\nu\nu$	0.129	$\nu\nu\nu$	0.128	
		$\nu e\mu$	0.125	$\nu e\mu$	0.125	
		$\nu\mu\mu$	0.072	$\nu\mu\mu$	0.073	
		$\boldsymbol{\mu\pi}$	**0.071**	$\boldsymbol{\mu\pi}$	**0.060**	
		$\nu\pi^0$	0.038	$\nu\pi^0$	0.032	
		Other	0.378	Other	0.419	
Total FASER2 $N \to \mu\pi$ Events			8610		81.8	10.6

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Statistical Methods

Maximum Likelihood

- We perform an **asymptotic likelihood-based test** to obtain the expected constraints on the HNL model parameters using Wilk's Theorem.
- **Nuisance parameters** are incorporated to represent the theoretical uncertainties in **forward hadron production.**
- We consider the following scenarios:
  - $\sigma_\eta = 60\,\%$ and $\sigma_\gamma = 10\,\%$ (Current)
  - $\sigma_\eta = 5\,\%$ and negligible σ_γ (Future)

Normalization Uncertainty

$$L(\vec{d}, m_N; \eta, \vec{\gamma}) = \prod_i f_P(d_i \,|\, \eta\gamma_i\mu_i)\underline{f_G(1\,|\,\eta, \sigma_\eta)}\,\underline{f_G(1\,|\,\gamma, \sigma_\gamma \oplus \sigma_i^{MC})}$$

Bin-by-bin Uncertainty (Shape)

$$\tilde{t}(\vec{d}\,|\,m_N, U_\mu) := -2\ln\frac{L(\vec{d}\,|\,m_N, U_\mu;\, \hat{\hat{\eta}}, \hat{\hat{\vec{\gamma}}})}{L(\vec{d}\,|\,\hat{m}_N, \hat{U}_\mu;\, \hat{\eta}, \hat{\vec{\gamma}})}$$

- d_i : Observed bin-count
- $\mu_i(m_N, U_\mu)$: Mean bin count prediction
- Nuisance Parameters: η, γ_i
- Theoretical Uncertainties:
  - σ_η : Normalization (60% , 5%)
  - σ_γ : Uncorrelated "Shape" (10% , 0%)
  - σ_i^{MC} : MC Statistics

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Kinematic Measurements

$$\boxed{m_{\mu\pi}^2 = m_\mu^2 + m_\pi^2 + 2(E_\mu E_\pi - p_\mu p_\pi \cos\theta_{\mu\pi})}$$

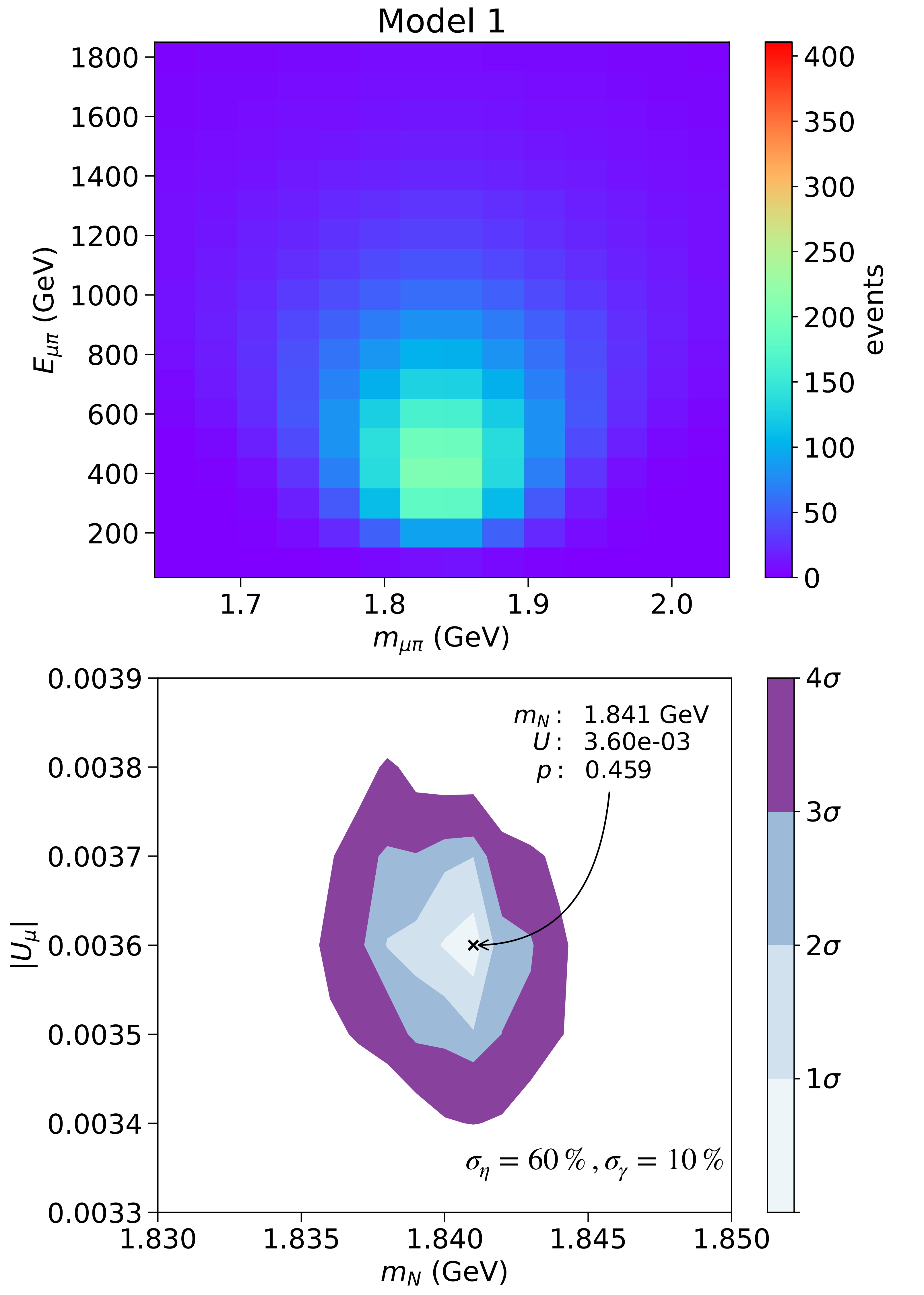
- Dual-readout hadronic calorimeter
  - Combines scintillation and Cherenkov signals for high precision measurement of the pion energy

$$\frac{\delta E_\pi}{E_\pi} = \frac{35\,\%}{\sqrt{E_{\pi^\pm}/\mathrm{GeV}}} \oplus 0.5\,\%$$

- Scintillating Fiber Tracking Stations
  - Measure muon momentum and angular separation

$$\frac{\delta p_\mu}{p_\mu} = 3.3\times 10^{-5}\left[\frac{p_\mu}{\mathrm{GeV}}\right], \quad \delta\theta_{\mu\pi} = 250\ \mu\mathrm{rad}$$

Model 1 (~8000 events)



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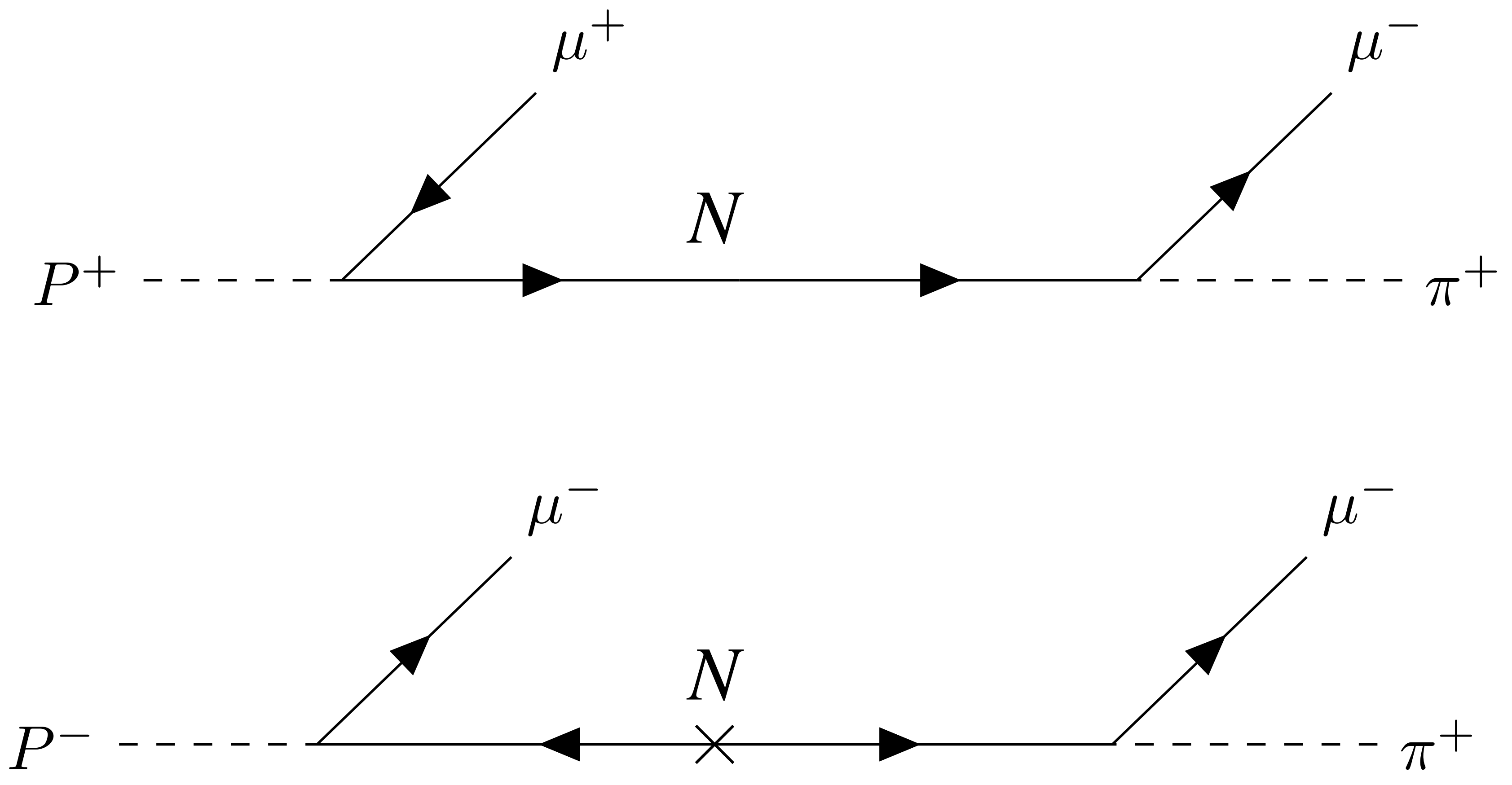
Majorana vs. Dirac HNLs

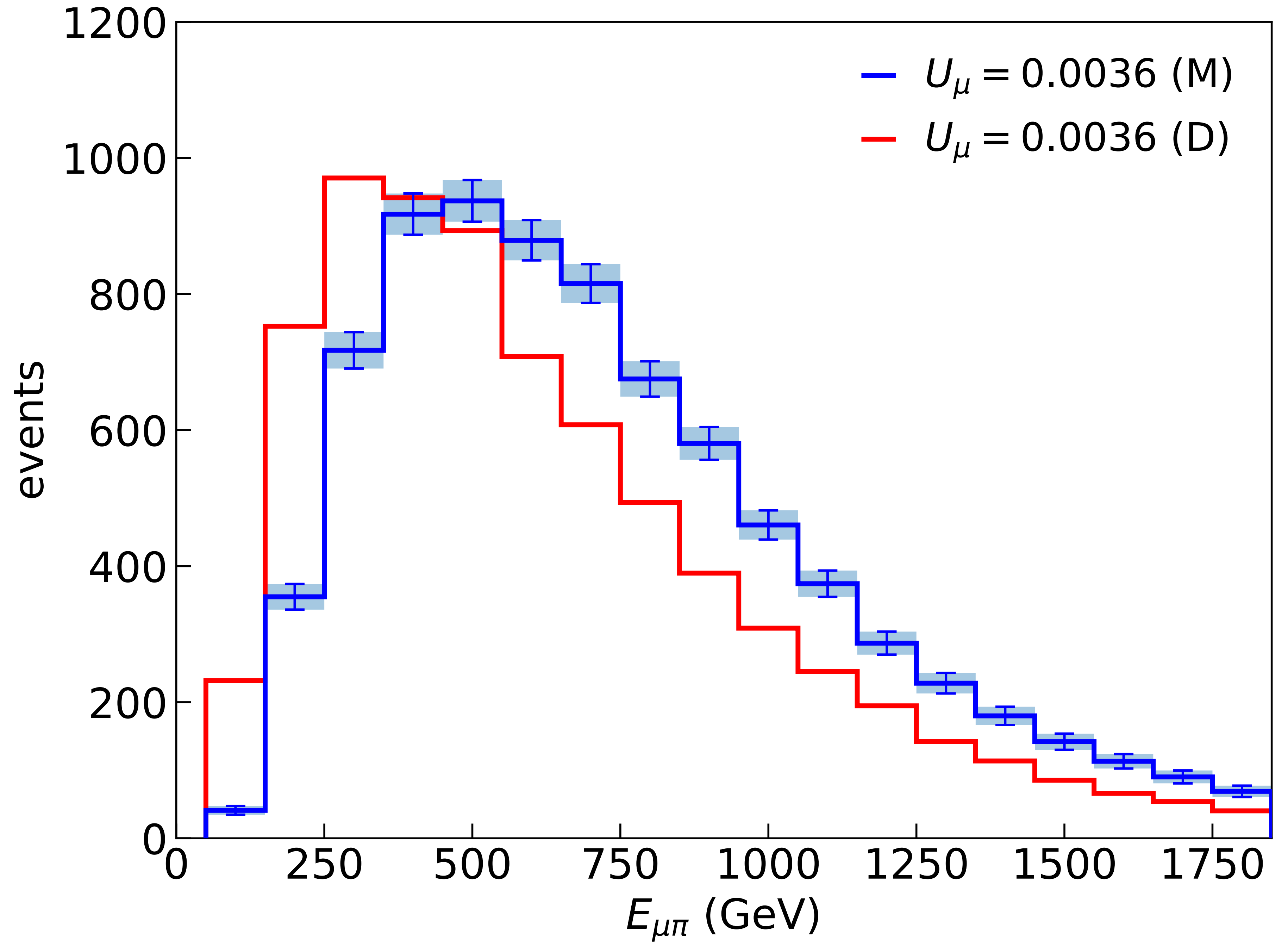
- Since SM neutrinos are massless, helicity and chirality eigenstates coincide, causing LNV processes involving SM neutrino to be suppressed by

$$\left|M\right|^2 \sim \left(\frac{m_\nu}{E_\nu}\right)^2$$

- However, for on-shell HNL decay, the mass dependence of the amplitude exactly cancels, resulting in **LNC and LNV processes which are equal and unsuppressed**
- Therefore, the lifetimes between Majorana and Dirac HNLs are related by

$$\tau_M = \frac{1}{\Gamma_{LNC} + \Gamma_{LNV}} = \frac{\tau_D}{2}$$





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Discrimination via Kinematics

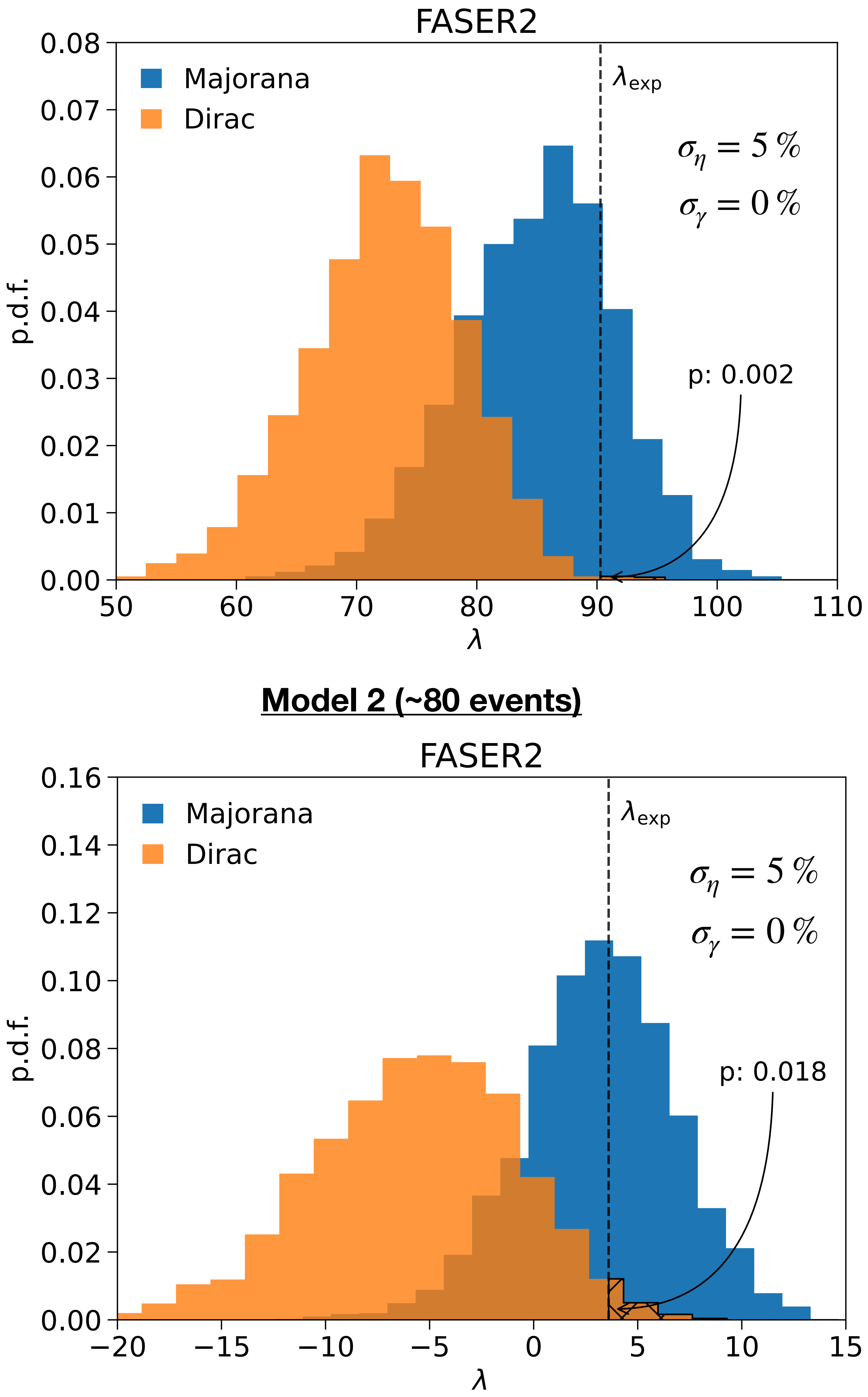
Profiled Likelihood Ratio Test

$$\lambda(\vec{d}\,) := -2\ln\frac{L_D(\vec{d}\mid \hat{m}_N^D, \hat{U}_\mu^D;\, \hat{\eta}^D, \hat{\vec{\gamma}}^D)}{L_M(\vec{d}\mid \hat{m}_N^M, \hat{U}_\mu^M;\, \hat{\eta}^M, \hat{\vec{\gamma}}^M)}$$

- Using λ, we compute the probability that a Dirac HNL would fluctuate to produce a signal that looks just as (or more) "Majorana-like" than the expected observation
- In the high statistics scenario **(Model 1)**, we find that FASER2 can expect to reject the Dirac hypothesis with $99.8\,\%$ CL (2.9σ)
- In the low statistics scenario **(Model 2)**, we find that FASER2 would only reject the Dirac hypothesis with $98.2\,\%$ CL **(**2.1σ**)**

Model 1 (~8000 events)

Model 2 (~80 events)



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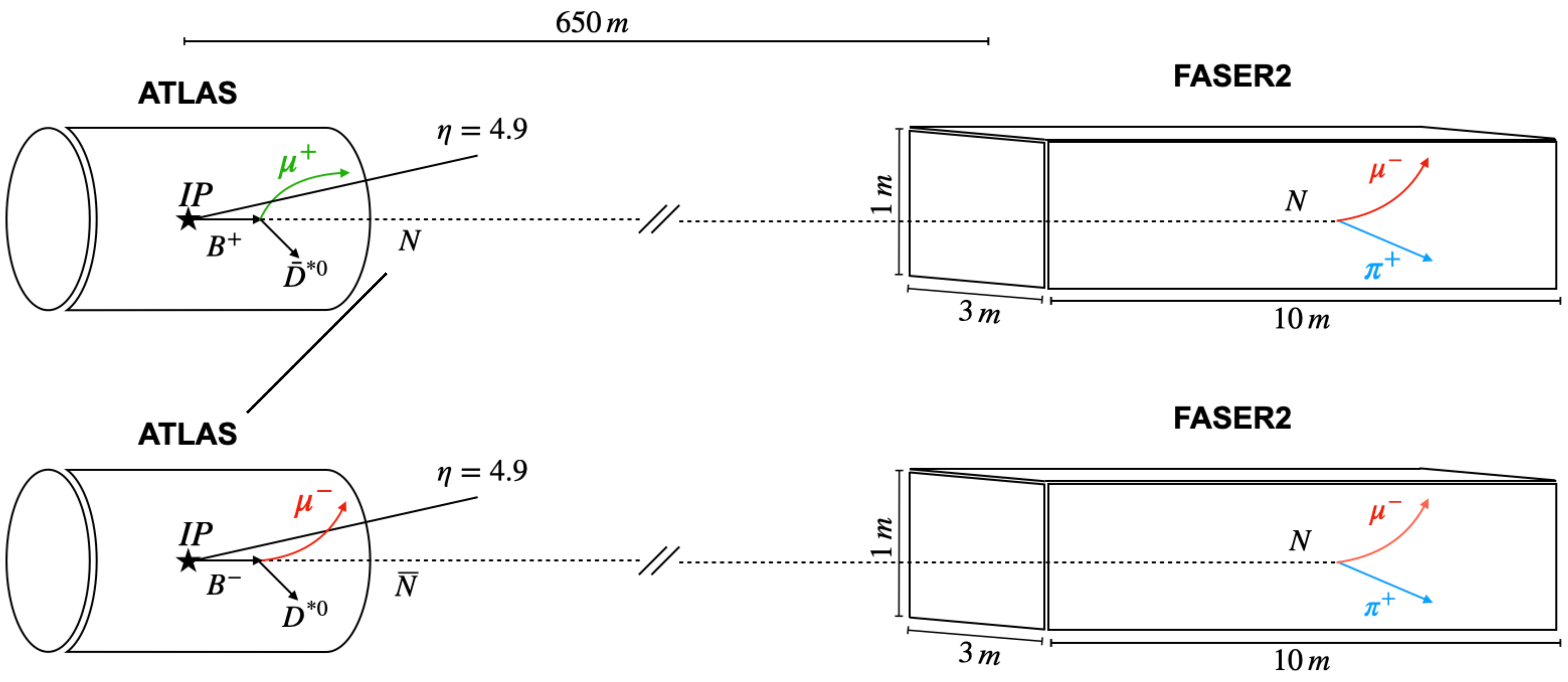
FASER2 as a Trigger for ATLAS

- In our work, we explore the idea of using **FASER2 as a trigger for ATLAS** to measure muons produced in association with HNLs as a probe of LNV.
    - ATLAS L0 Latency: $6\,\mu s$
    - Triggering Time: $2.2\,\mu s + 2.9\mu s$
- In the absence of background, one could determine LNV on an event-by-event basis. **However this is not in the absence of background!**



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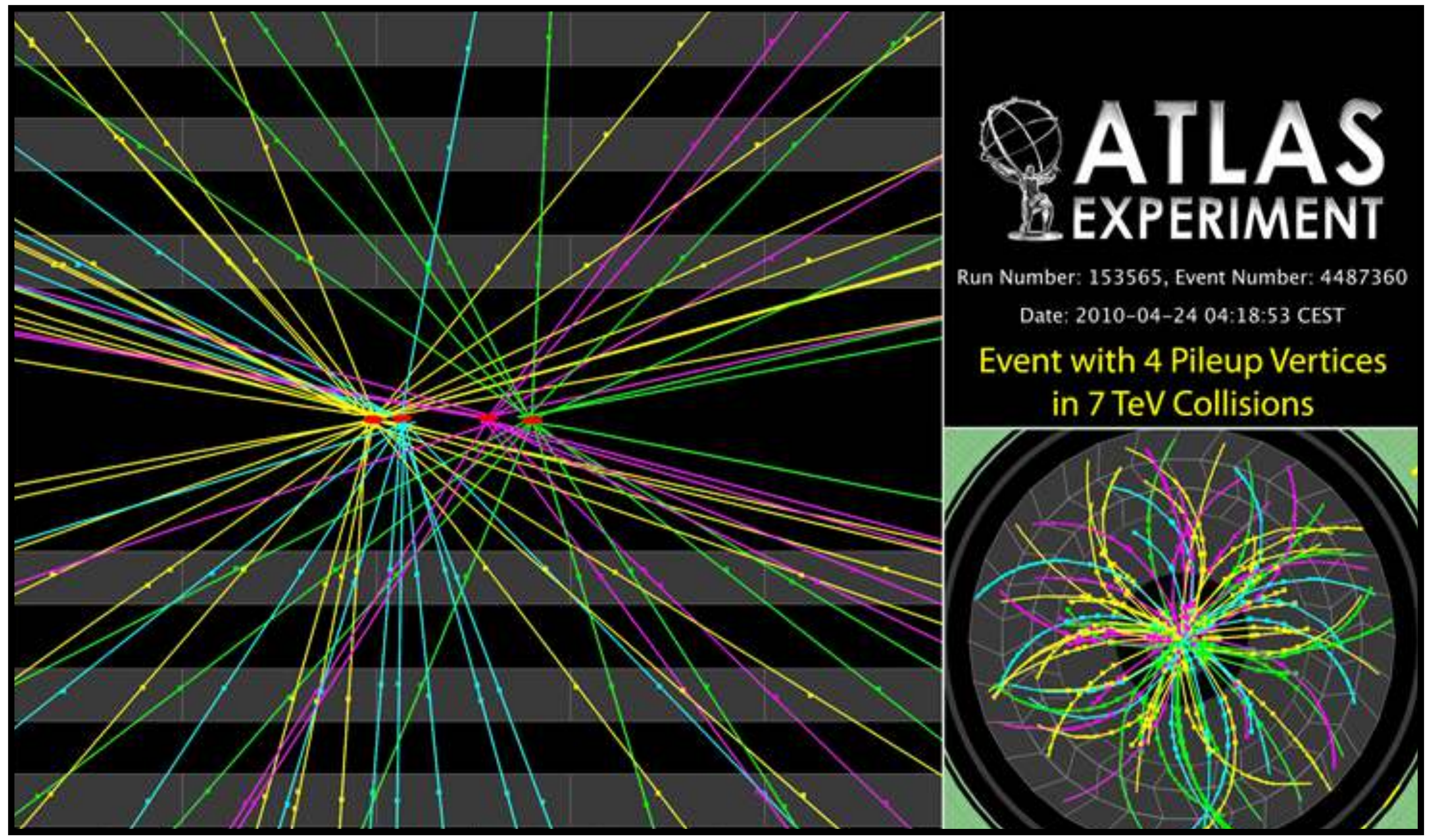
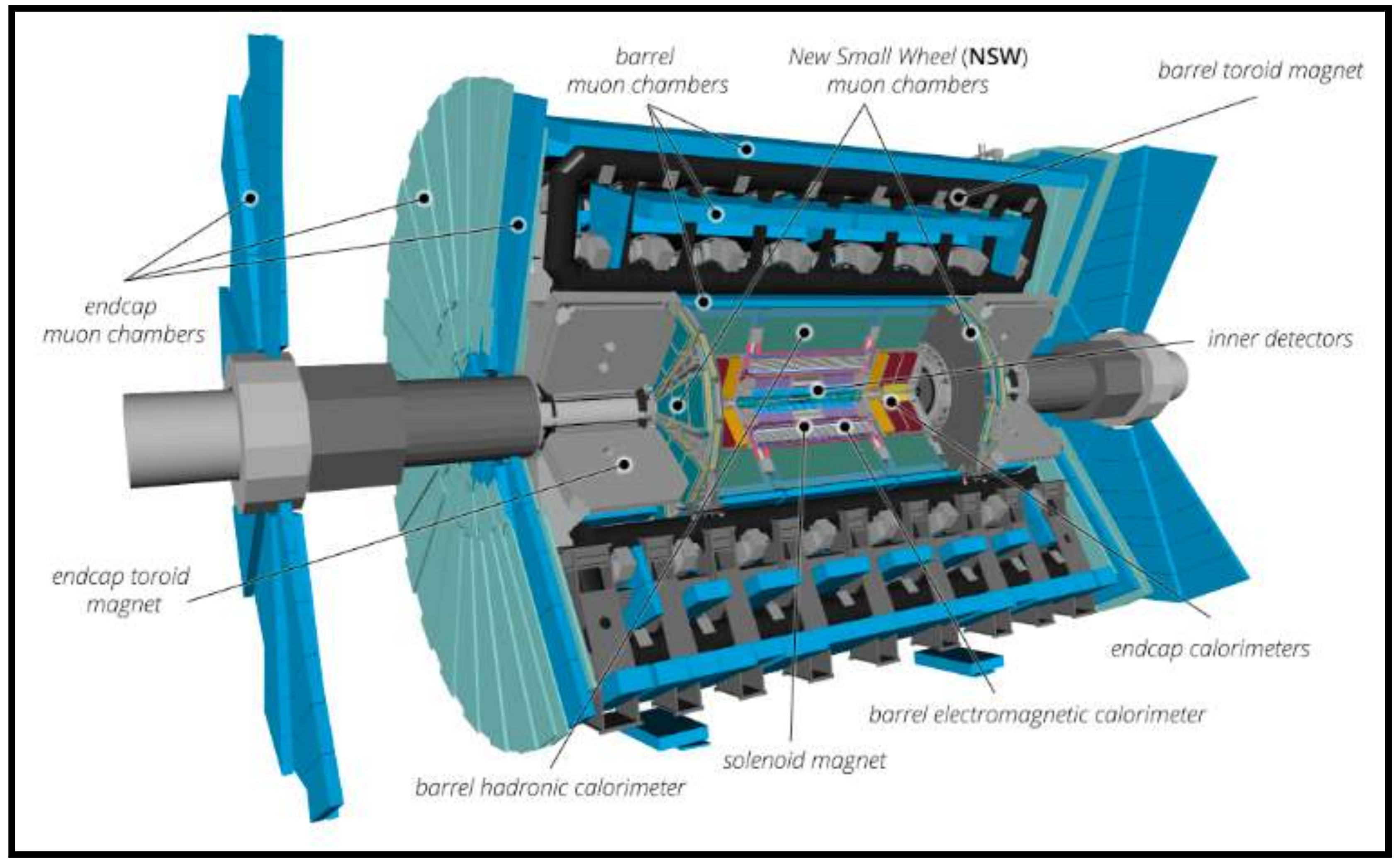
Pileup Background

- At the HL-LHC, there are $\sim 10^{11}$ protons per bunch, resulting in $\langle \mu \rangle = 200$ pile-up events per crossing.
- We use Pythia8 to overlay 200 pp collisions to generate an estimate for the number of background tracks per bunch crossing
- A high-η muon tagger is required to reduce background from charged tracks
  - Proposed in ATLAS Phase-II upgrade (ATLAS Collaboration 2015)



[https://lhc-closer.es/taking_a_closer_look_at_lhc/0.pile_up]



[ATLAS Collaboration 2023]

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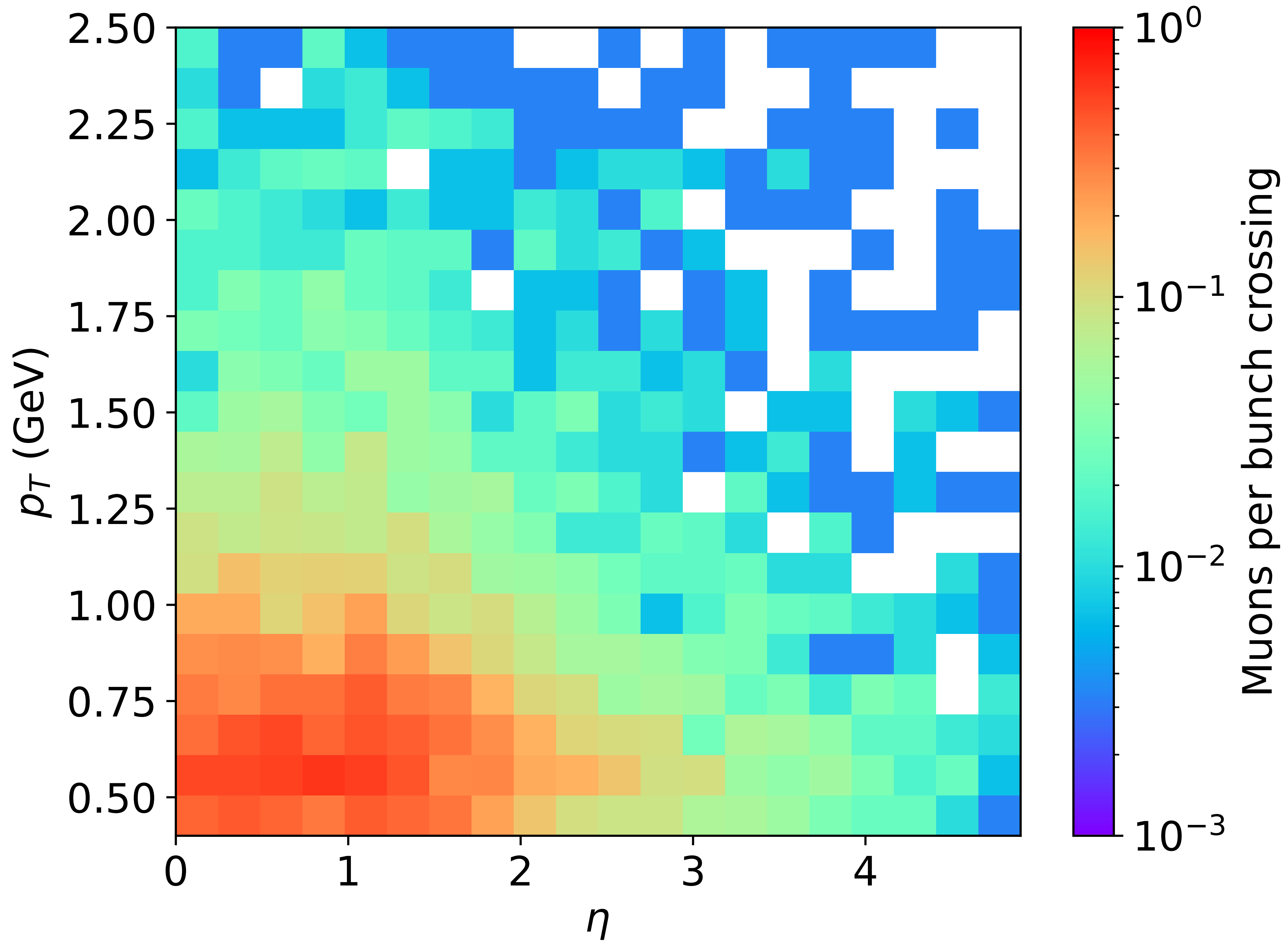
- We use Pythia8 to overlay 200 pp collisions to generate an estimate for the number of background tracks per bunch crossing
- A high-η muon tagger is required to reduce background from charged tracks
  - Proposed in ATLAS Phase-II upgrade (ATLAS Collaboration 2015)
- We find that there is **roughly 1 uncorrelated background muon per bunch crossing.**



Parent	Muons ($\lvert\eta\rvert < 4.9$)	Muons ($3.5 < \eta < 4.9$)
$\pi^\pm$	31.41	0.43
$K^\pm$	16.09	0.26
K_L^0	0.37	0.0033
Heavy Mesons	3.34	0.33

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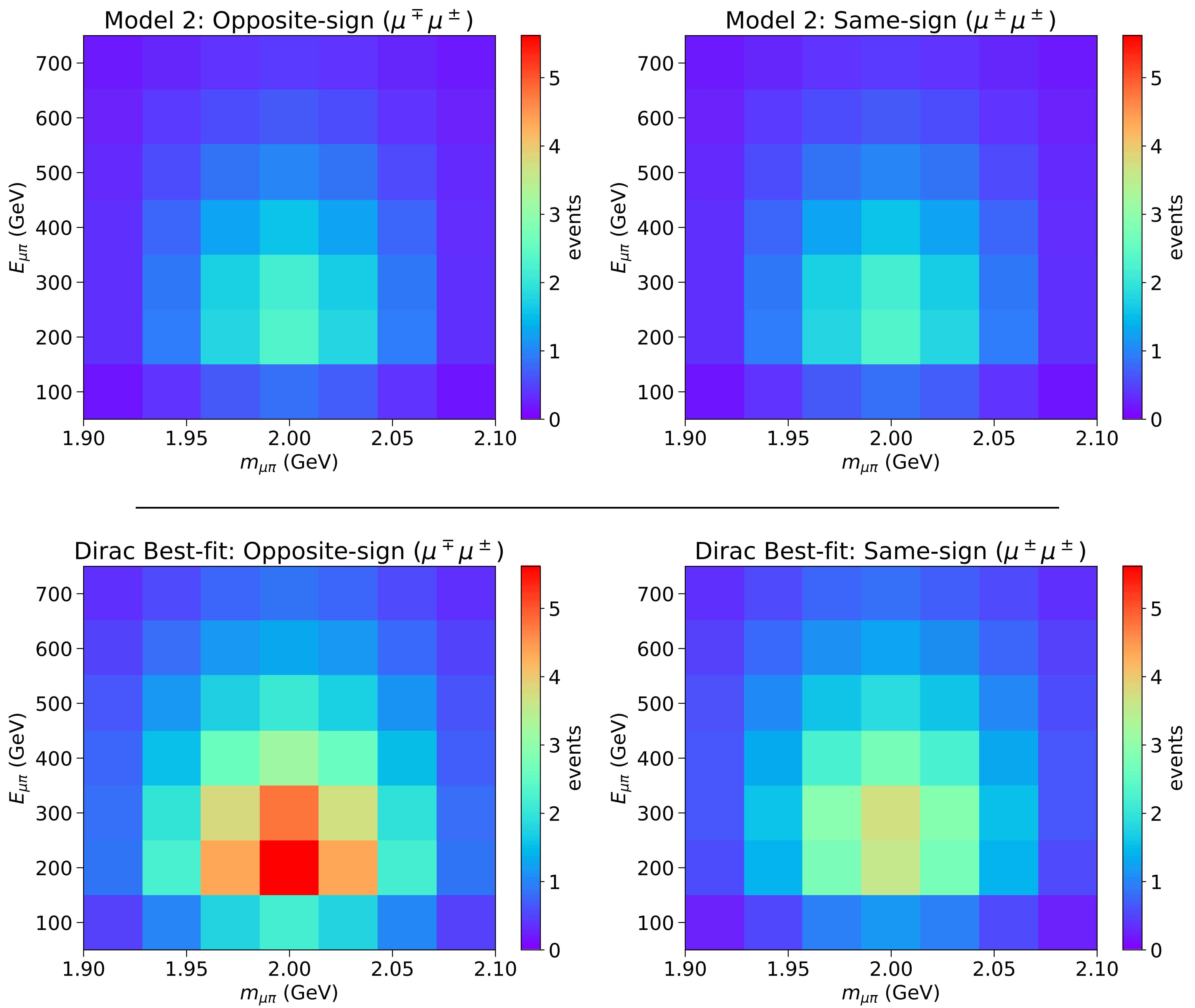
Correlated Signal

$$\mu_i^{SS} = \frac{b}{2}\,\mu_i^F + \frac{\eta_M}{2}\,\mu A_i$$

$$\mu_i^{OS} = \frac{b}{2}\,\mu_i^F + \left(1 - \frac{\eta_M}{2}\right)\,\mu_i^A$$

$$\eta_M = \begin{cases} 1 & \text{Majorana} \\ 0 & \text{Dirac} \end{cases}$$

- We separate each triggered HNL event into the number of muons detected in ATLAS that have either the **same sign (SS)** or **opposite sign (OS)** as the muon detected in FASER2.
- Muons produced in association with **Majorana HNL's** can be found with **OS or SS muons** in FASER2 with equal rates, resulting in an **symmetric distribution.**
- In contrast, **Dirac HNL's** can only be found with **OS muons** in FASER2, resulting in an **asymmetric distribution.**



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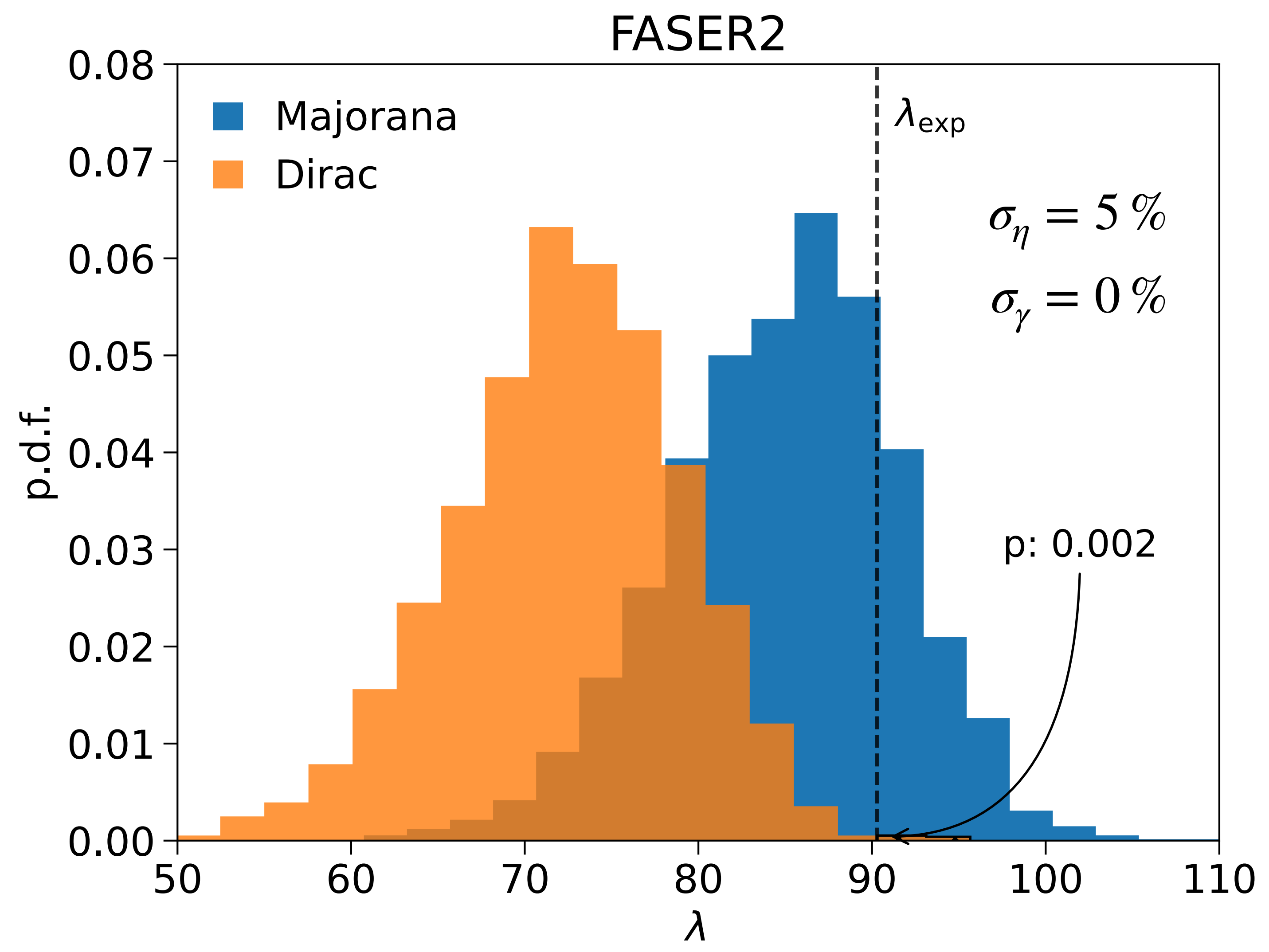
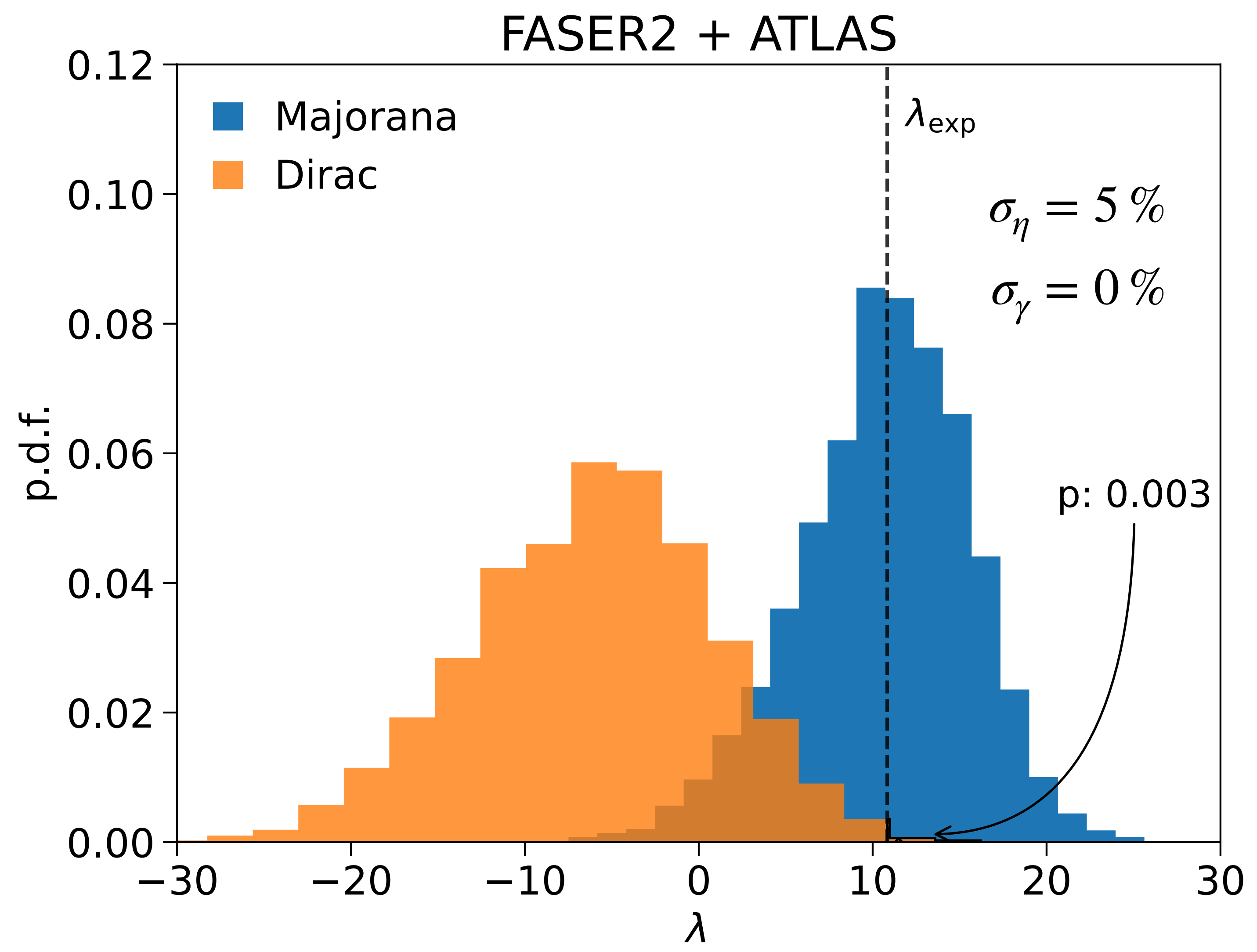
FASER2 as a Trigger for ATLAS

- Using the **correlated distributions**, we find that FASER2 can expect to reject the Dirac hypothesis with $99.7\,\%$ CL (2.7σ), **despite having significantly lower statistics!**
- This illustrates that the main LHC detectors and proposed auxiliary detectors can have a synergistic relationship on an event-by-event basis.

Model 1 (~8000 events)



Model 2 (~80 events)



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Conclusion

- Heavy Neutral Leptons are a simple, well-motivated extension to the SM which have a rich phenomenology to be studied at colliders
- We have presented the capability of FASER2 to:
  - Measure and constrain the mass and coupling of an HNL in the event of a signal detection
  - Differentiate Majorana and Dirac HNL's by studying the impact of their lifetimes on the energy distributions of HNLs
  - Act as a trigger for ATLAS to improve discrimination between Dirac and Majorana HNLs by studying the correlation of muons produced at ATLAS and detected at FASER2

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Backup Slides

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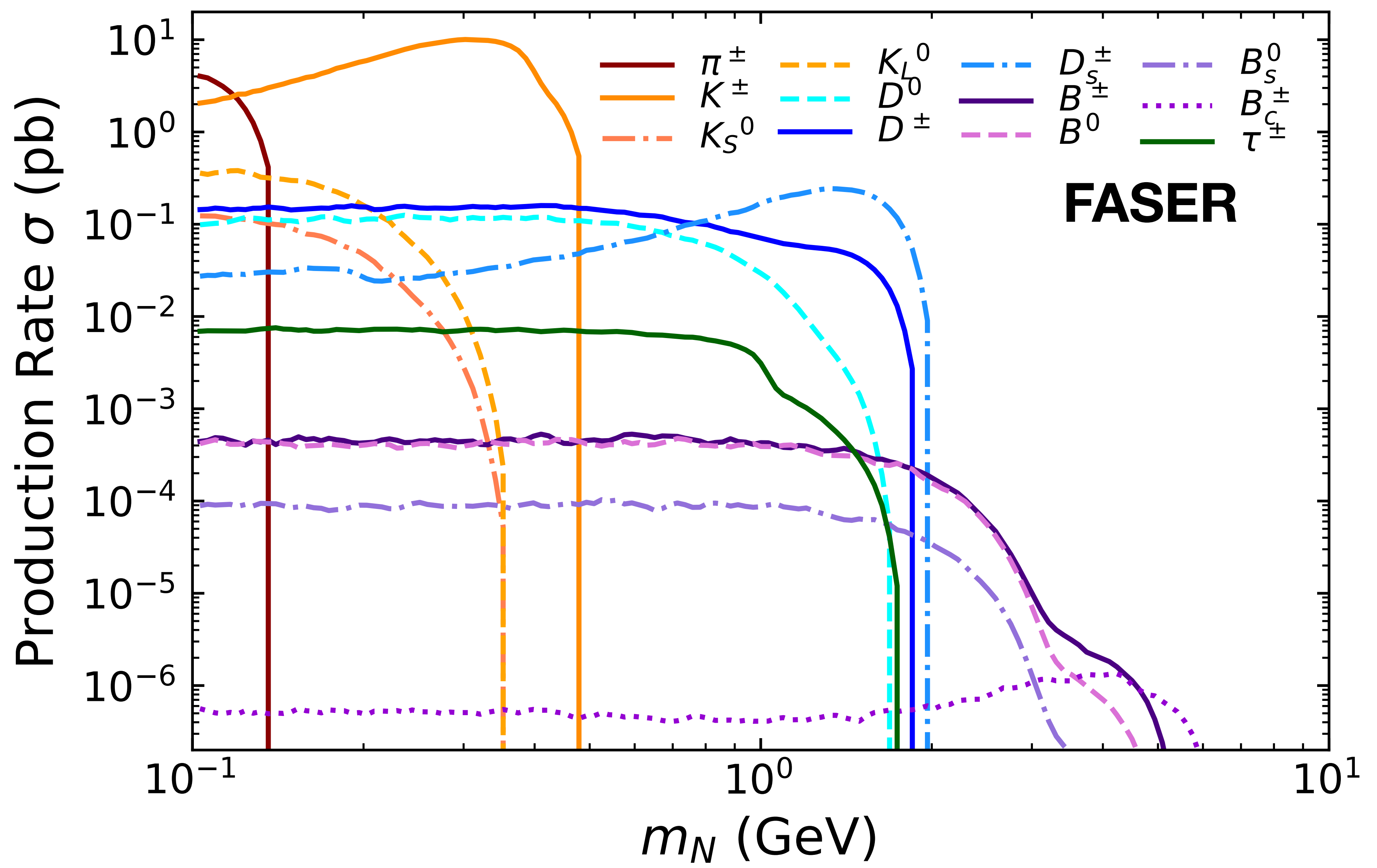
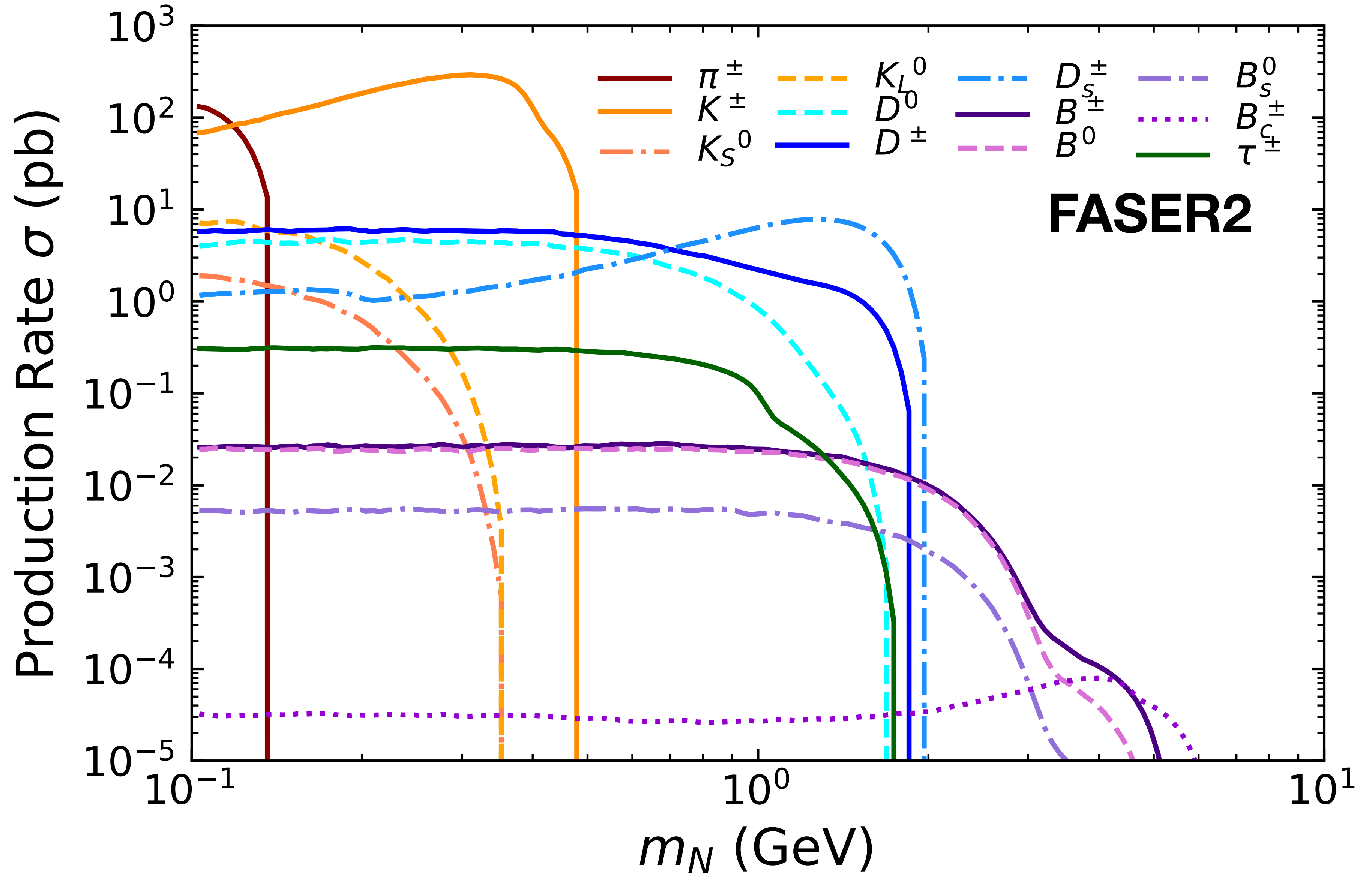
Hadron Production

- FORESEE generates HNL production rates through sampling of decays of parent particles, whose spectra are generated via MC event generators
- The primary source of uncertainty in HNL production rates is the modeling of the hadron fluxes
- For light hadrons (π, K), we estimate this uncertainty by considering the predictions of EPOS-LHC, QGSJET 2.04, and Sibyll 2.3d
- For heavy hadrons and taus, the uncertainty is estimated by considering predictions from POWHEG+Pythia with different choices of factorization scale





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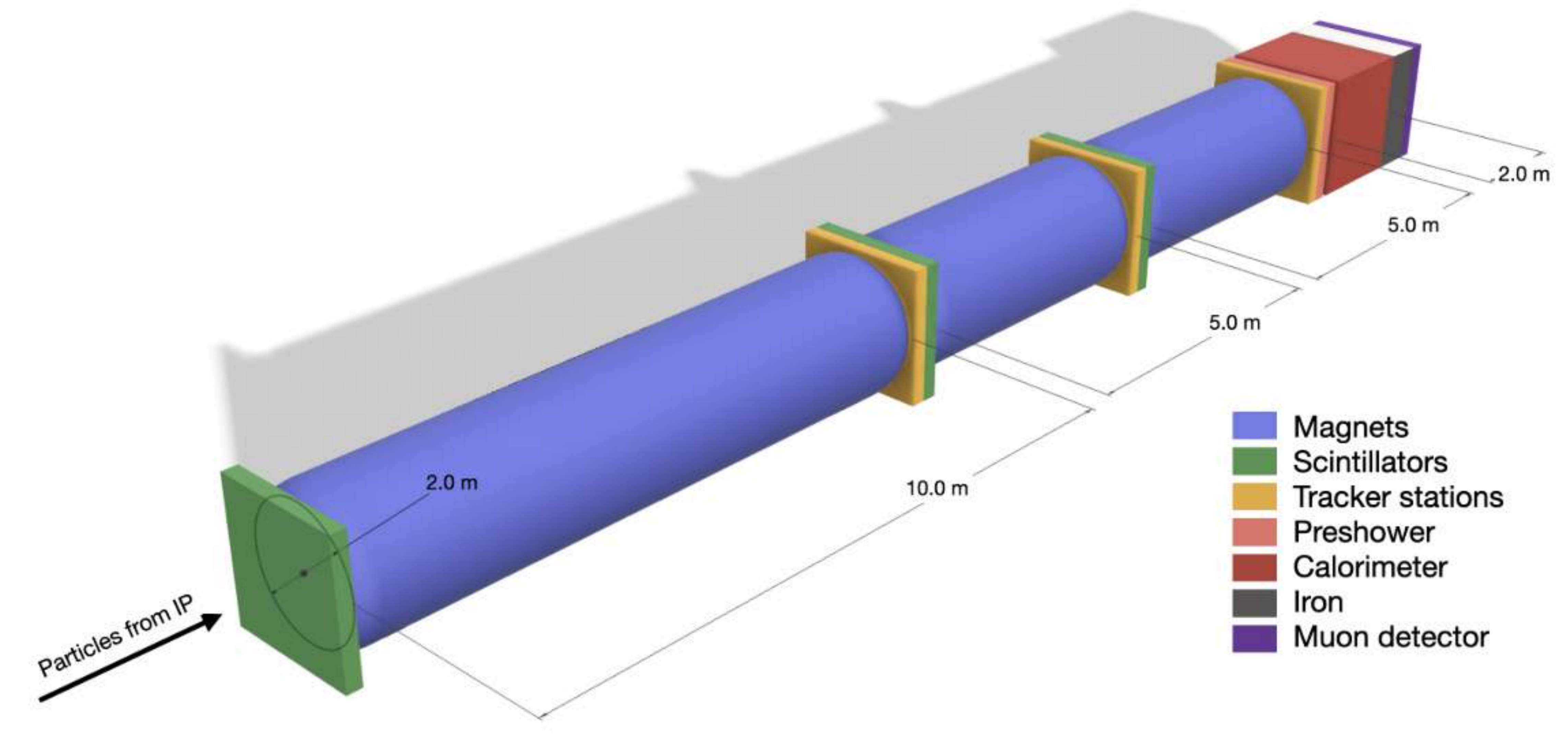
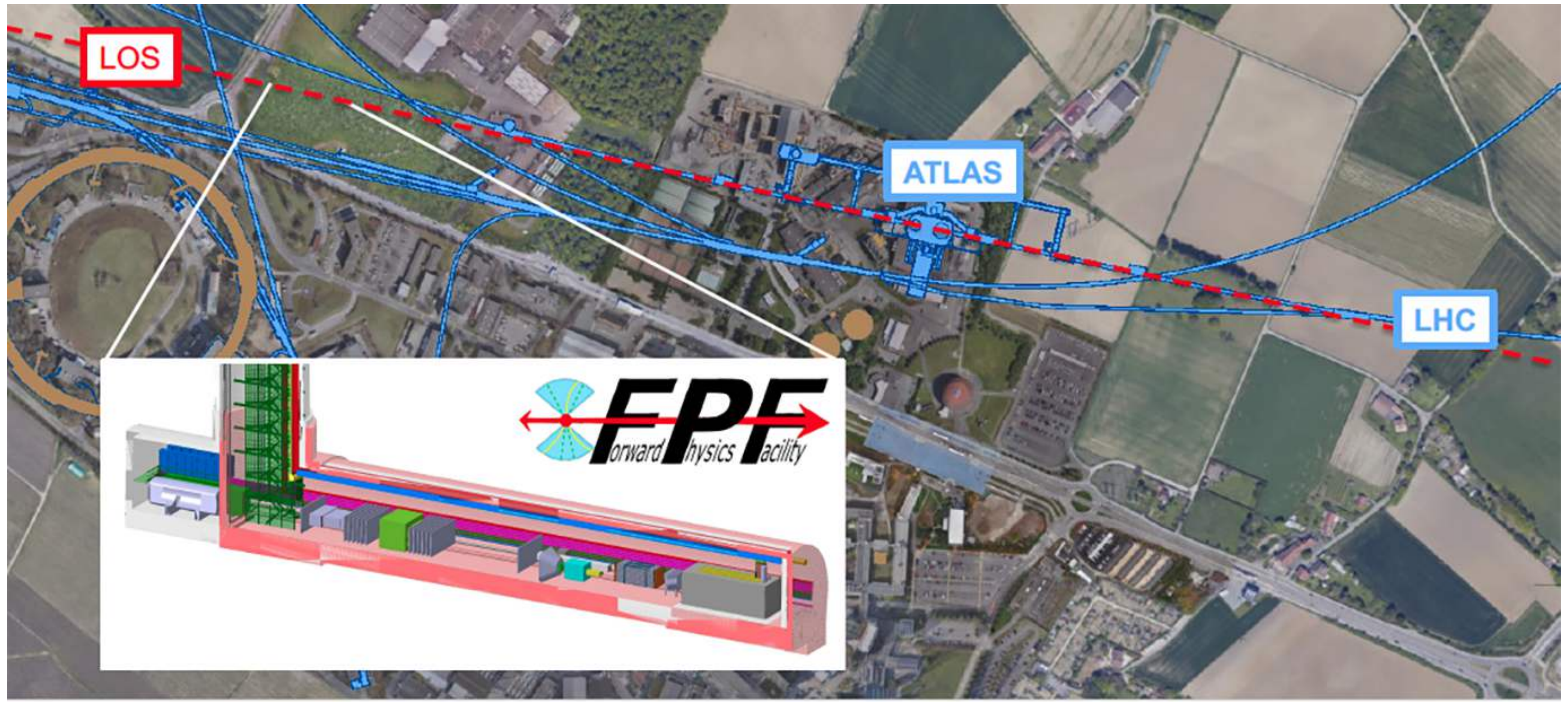
FPF: FASER2

- The Forward Physics Facility (FPF) is a proposed underground facility to house a suite of forward physics experiments during the High-Luminosity LHC era (HL-LHC)
- FASER2 is a proposed FPF experiment with an enlarged decay volume compared to FASER, which could extend the physics reach of the FASER program significantly
- Located 650 meters down the beam axis, with a rectangular decay volume with cross-section $3 \text{ m} \times 1 \text{ m}$ and length $\Delta = 10$ m.
- Angular coverage of $\theta \lesssim 2.4$ mrad $(\eta \gtrsim 6.7)$



[Feng et al. 2203.05090]



[https://fpf.web.cern.ch/]

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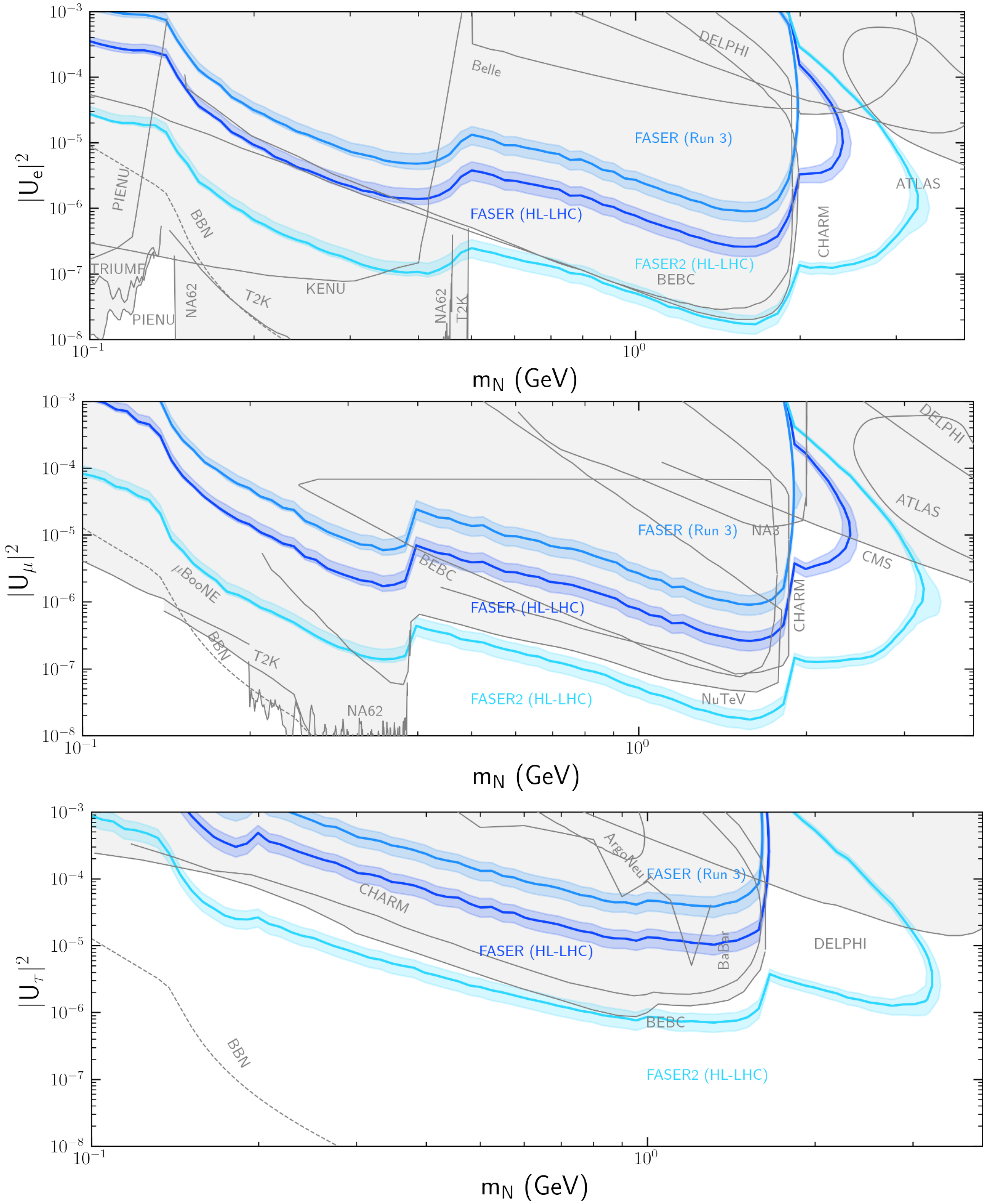
Results: FASER Sensitivity

- We present results for FASER during Run 3 ($\mathscr{L} = 250$ fb^{-1}), and FASER/FASER2 during HL-LHC ($\mathscr{L} = 3000$ fb^{-1})
- We find that the HL-LHC upgrade will enhance the reach of FASER/FASER2 into unconstrained parameter space, with FASER2 seeing a significant increase in sensitivity due to its larger decay volume
- Results are easily obtained for different coupling ratios using the FORESEE Module!



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		$\boldsymbol{\mu\pi}$	**0.071**	$\boldsymbol{\mu\pi}$	**0.060**	
		$\nu\pi^0$	0.038	$\nu\pi^0$	0.032	
		Other	0.378	Other	0.419	
Total FASER2 $N \to \mu\pi$ Events			8610		81.8	10.6

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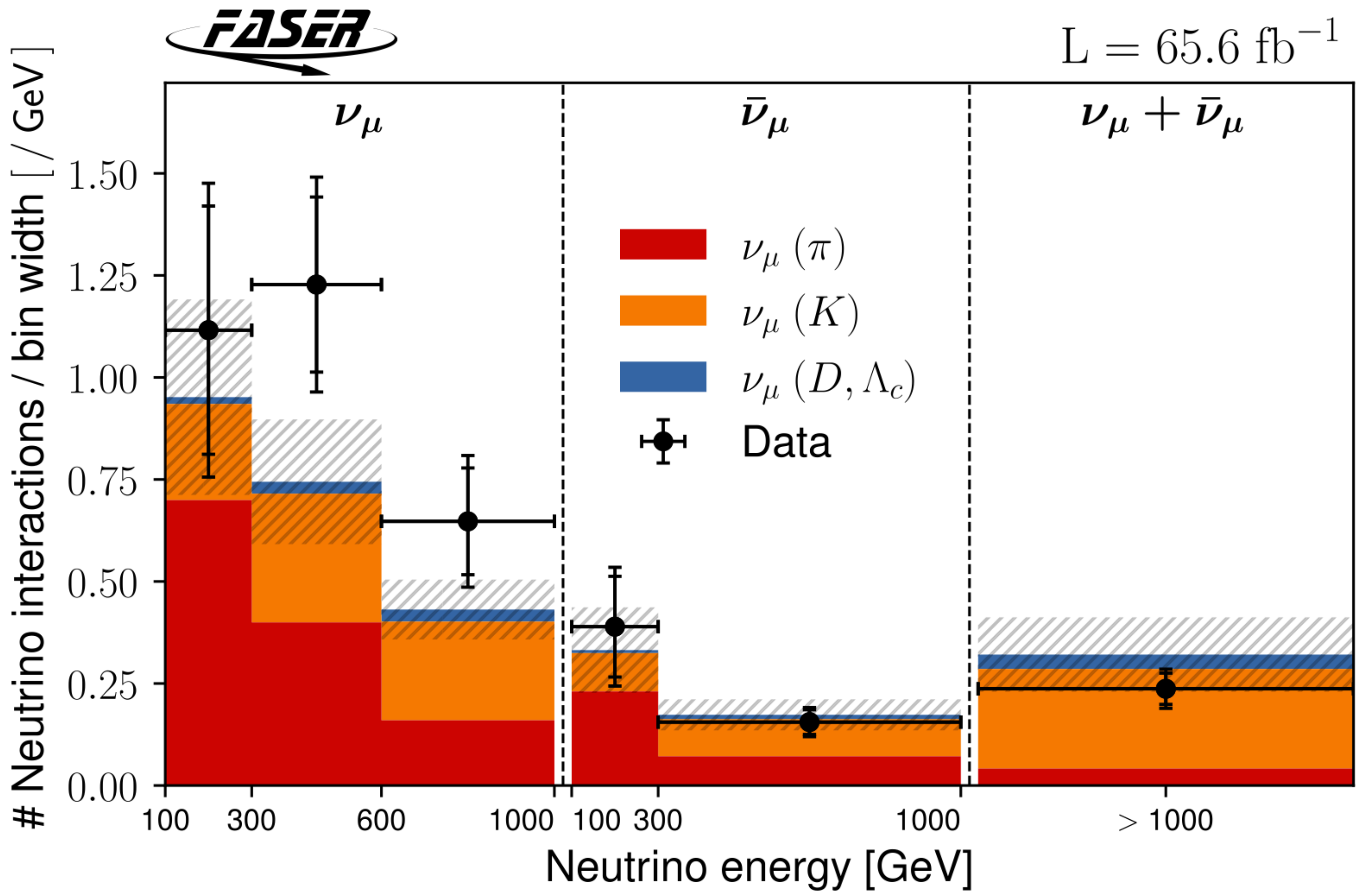
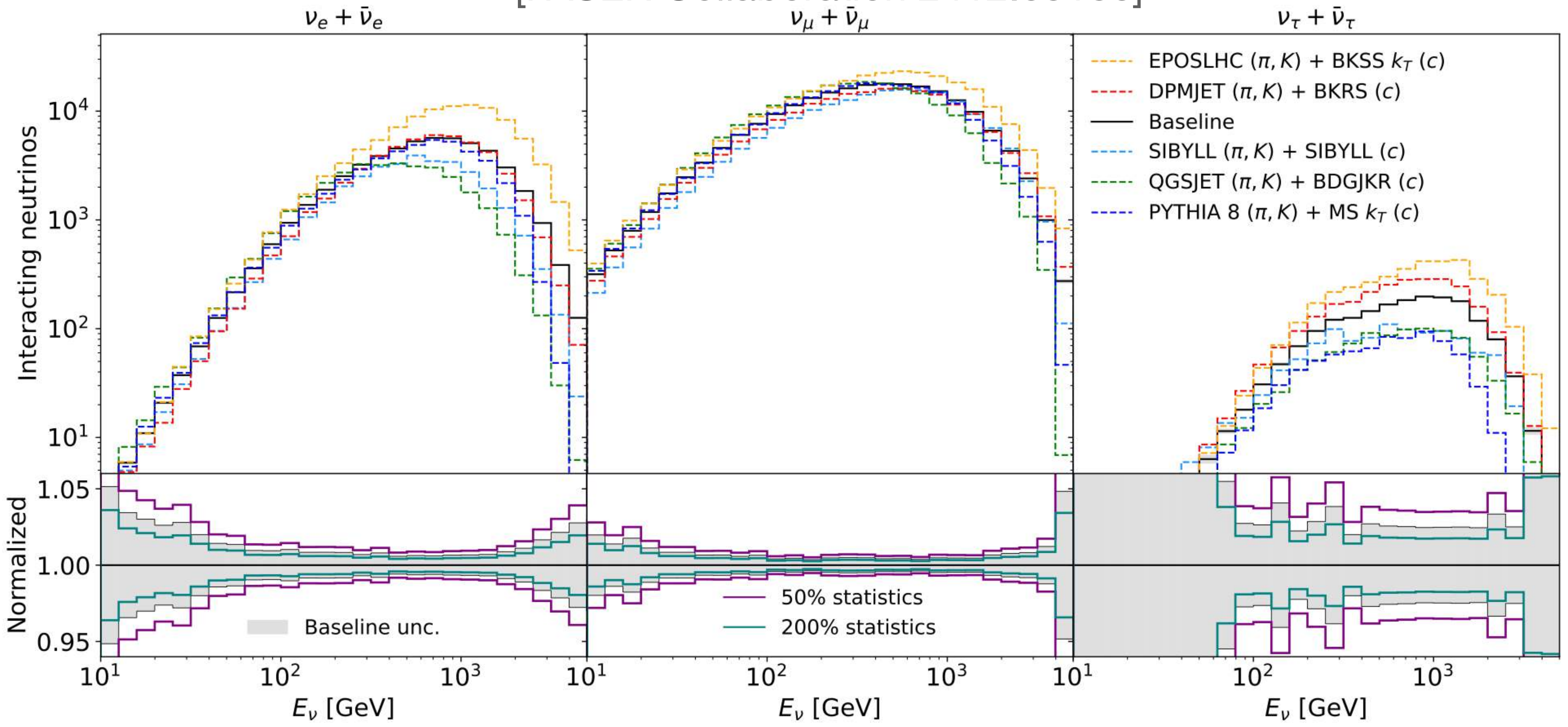
Theoretical Uncertainty

- The leading theoretical uncertainty in the HNL flux comes from **forward hadron production**
  - Hadron production in the forward direction at the LHC has not been measured historically
  - MC generators are not optimally tuned for the forward direction
- Fortunately, modeling uncertainties will certainly be reduced as FASERν continues to measure forward neutrino fluxes
  - *"Latest neutrino results from the FASER experiment and their implications for forward hadron production"* (Ohashi et al. 2025)



[FASER Collaboration 2412.03186]



[Kling, Makela, and Trojanowski 2309.10417]

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Parameter Estimation

$$L(\vec{d}, m_N; \eta, \vec{\gamma}) = \prod_i f_P(d_i \,|\, \eta\gamma_i\mu_i)\underbrace{f_G(1\,|\,\eta, \sigma_\eta)}_{\text{Normalization Uncertainty}}\underbrace{f_G(1\,|\,\gamma, \sigma_\gamma \oplus \sigma_i^{MC})}_{\text{Bin-by-bin Uncertainty (Shape)}}$$

Model 1 (High Statistics)

- **Assuming current flux uncertainties,** m_N & U_μ can be constrained to $0.1\,\%$ and $3\,\%$ respectively.
- **Assuming improved flux uncertainties,** m_N & U_μ can be constrained to $0.1\,\%$ and $2\,\%$ respectively.

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Parameter Estimation

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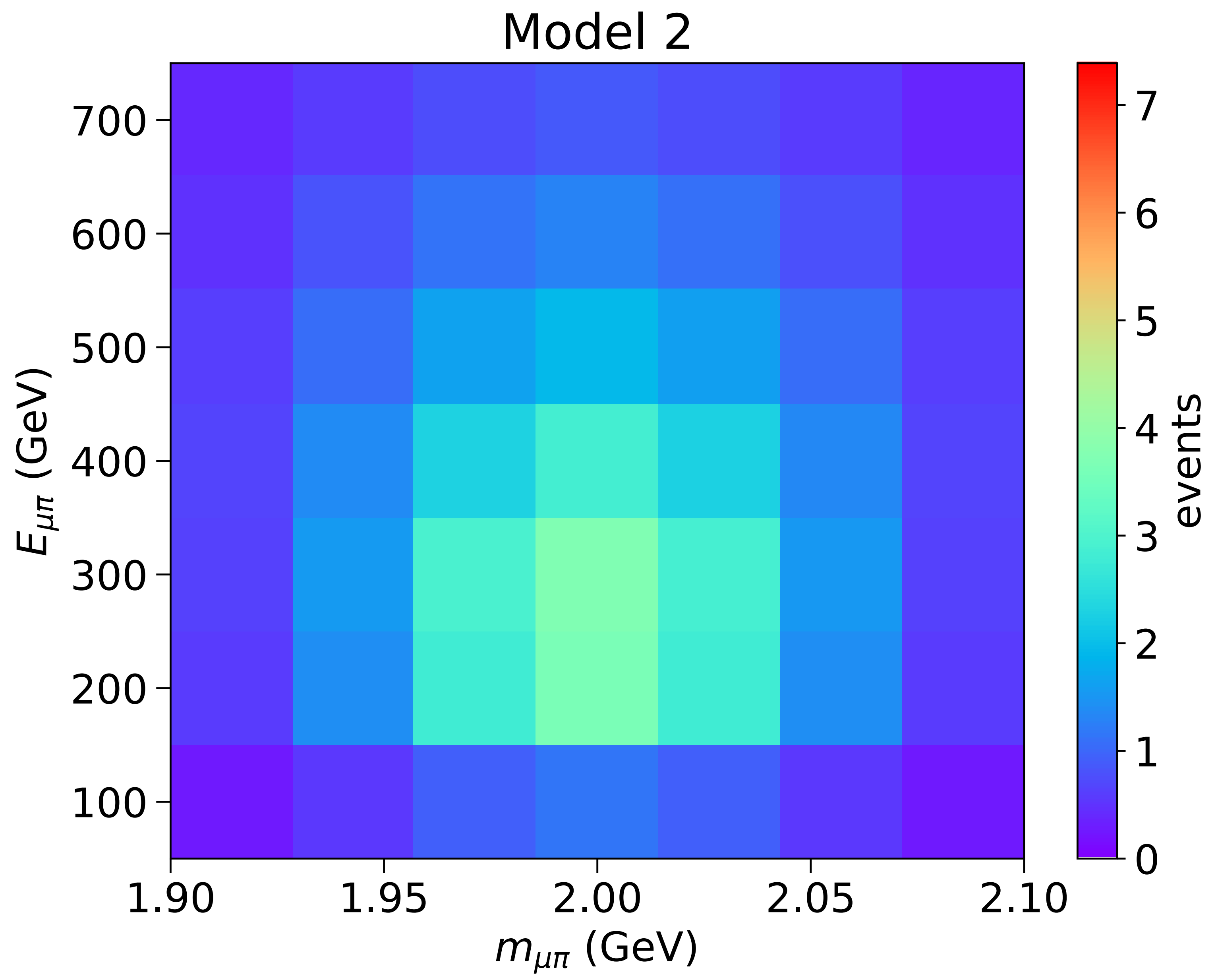
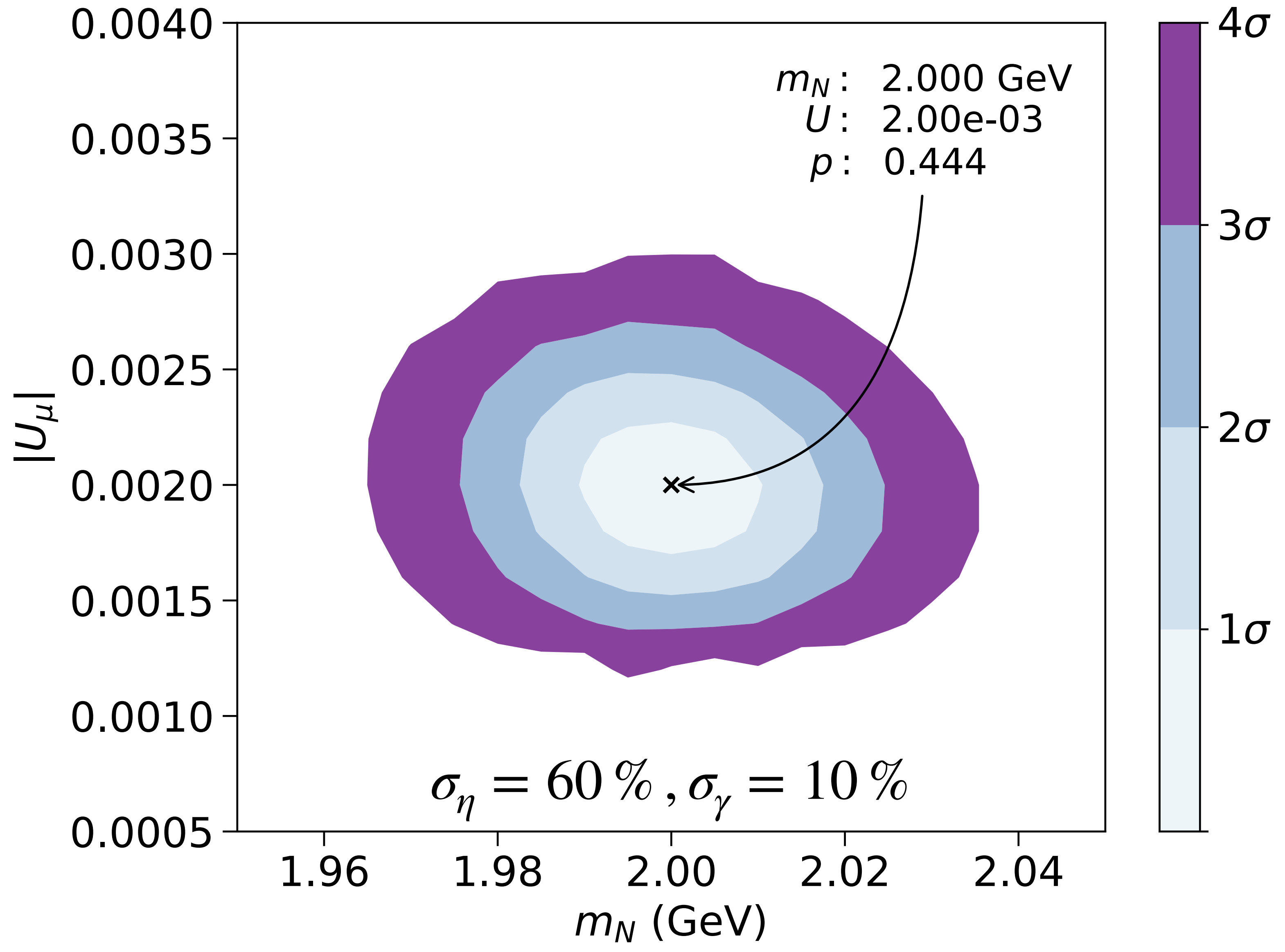
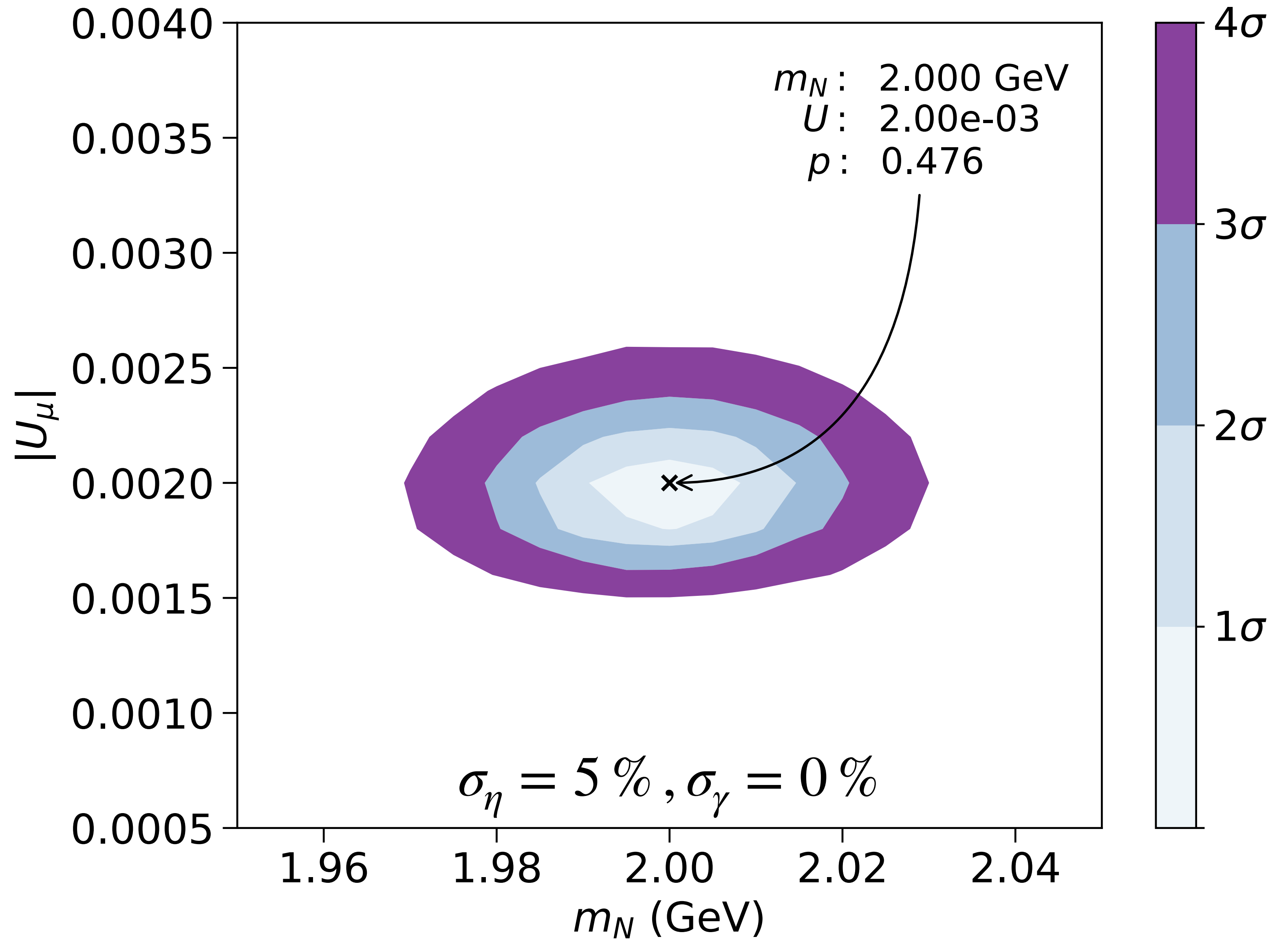
Model 2 (Low Statistics)







- **Assuming current flux uncertainties,** m_N & U_μ can be constrained to $1\,\%$ and $25\,\%$ respectively.
- **Assuming improved flux uncertainties,** m_N & U_μ can be constrained to $1\,\%$ and $15\,\%$ respectively.

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Majorana vs. Dirac HNLs

- Since SM neutrinos are massless, helicity and chirality eigenstates coincide, causing LNV processes involving SM neutrino to be suppressed by

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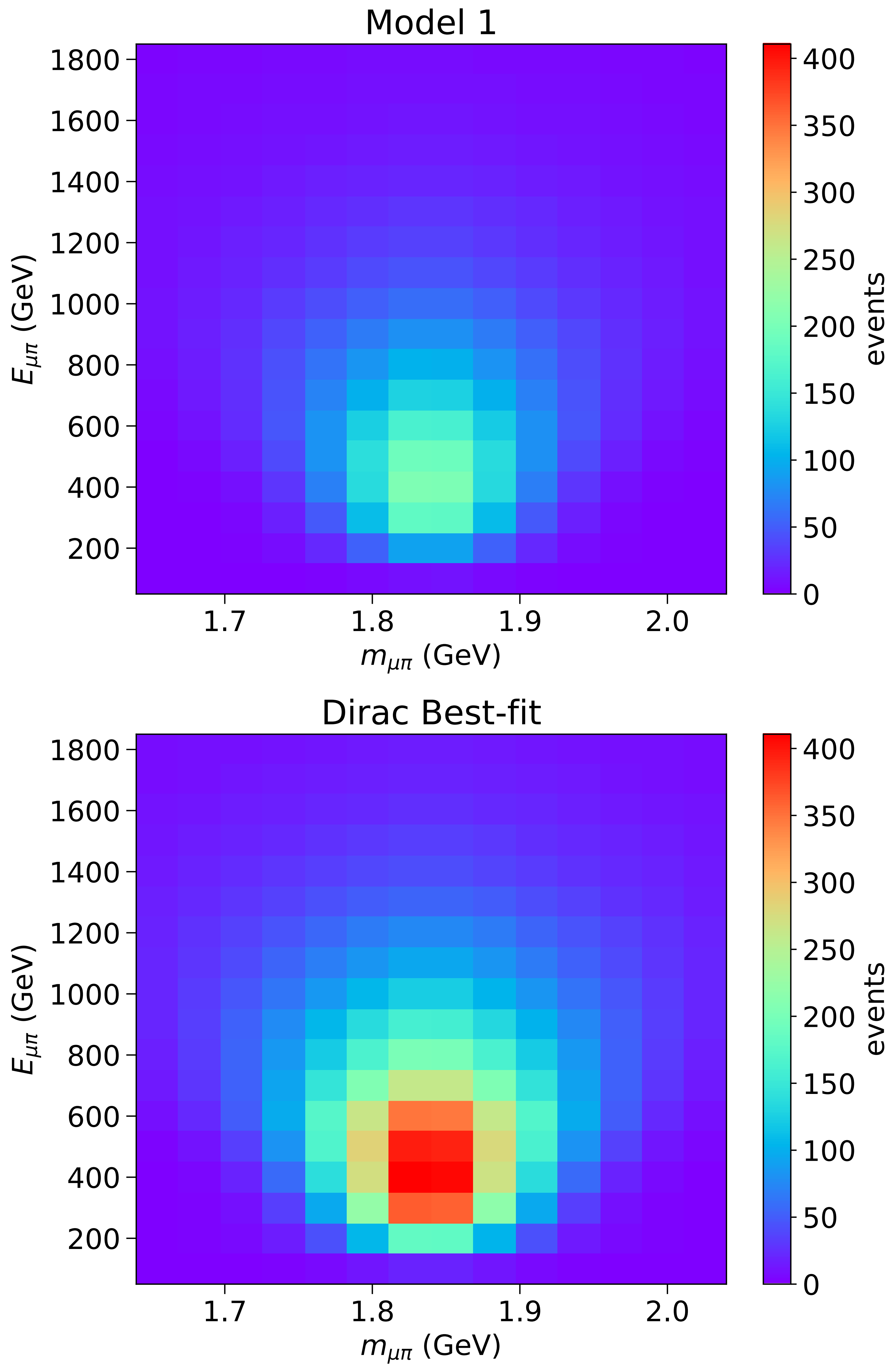

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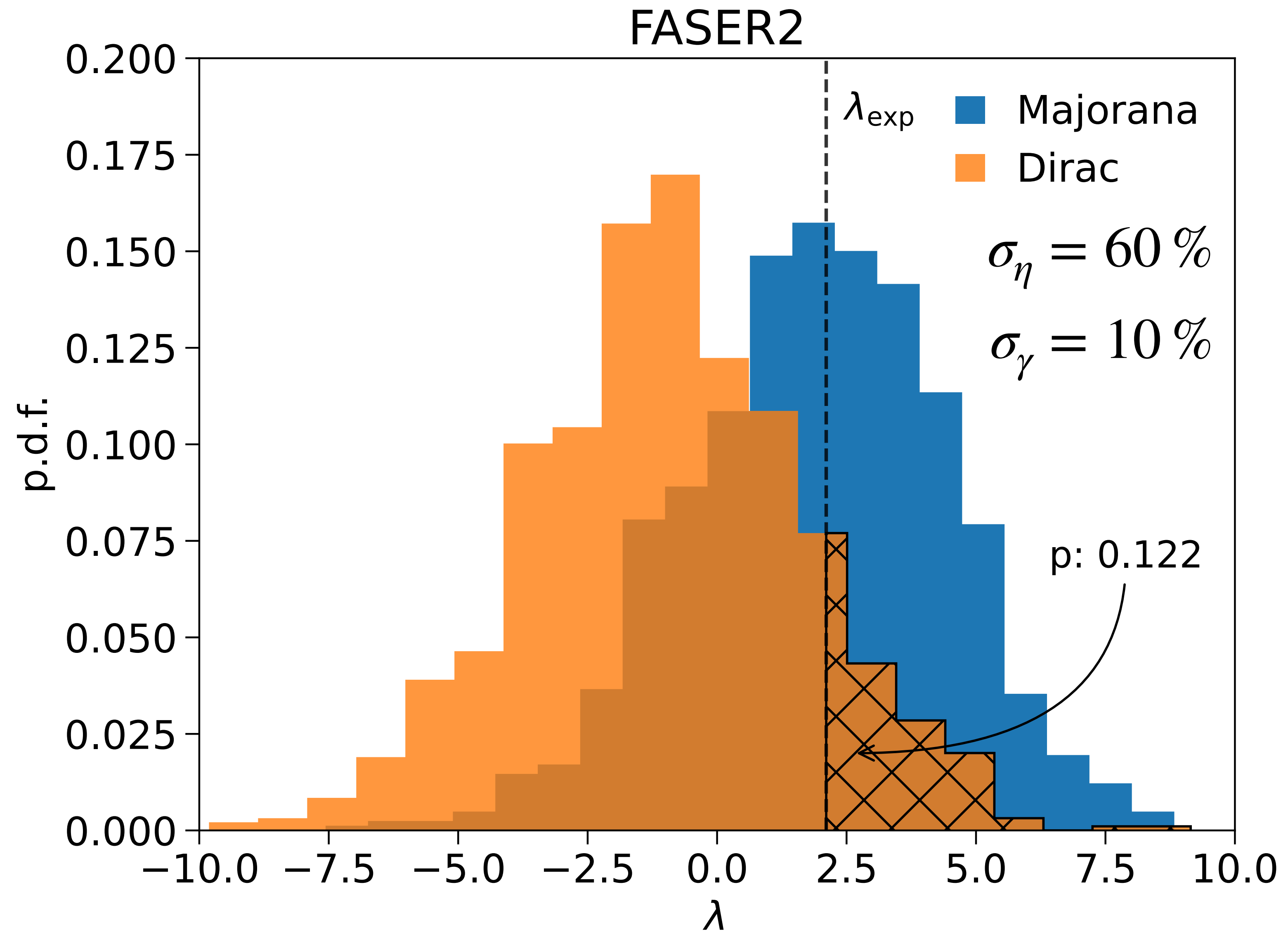
Discrimination via Kinematics

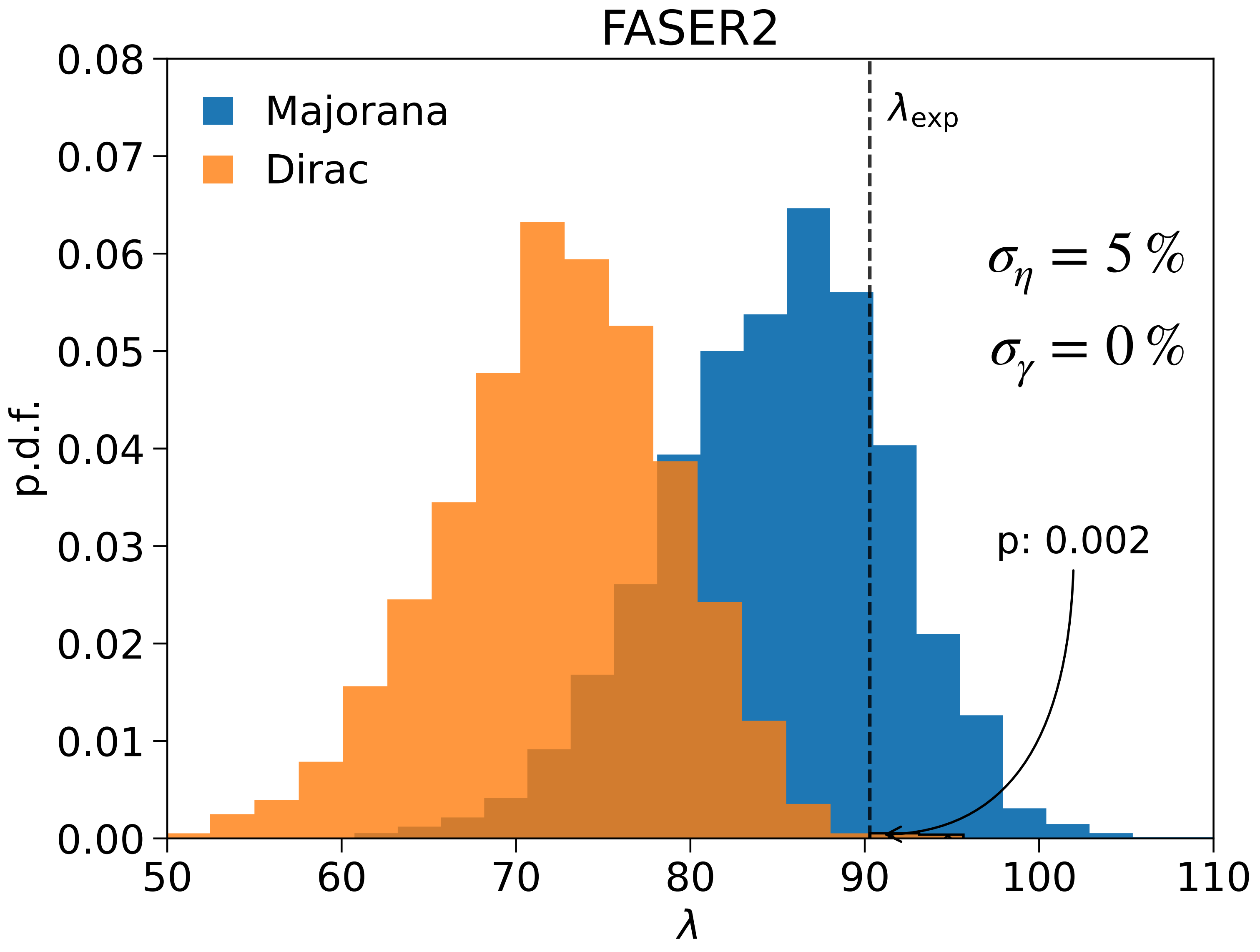
Model 1 (High Statistics)

Profiled Likelihood Ratio Test

$$\lambda(\vec{d}\,) := -\,2\ln\frac{L_D(\vec{d}\,|\,\hat{m}_N^D, \hat{U}_\mu^D;\, \hat{\eta}^D, \hat{\vec{\gamma}}^D)}{L_M(\vec{d}\,|\,\hat{m}_N^M, \hat{U}_\mu^M;\, \hat{\eta}^M, \hat{\vec{\gamma}}^M)}$$

- Using λ, we compute the probability that a Dirac HNL would fluctuate to produce a signal that looks just as (or more) "Majorana-like" than the expected observation
- Generated via $\mathcal{O}(10^3)$ pseudo-experiments, where the nuisance parameters for both models have been fixed to their best-fit values.
- Assuming flux uncertainties are improved, we find that FASER2 can expect to reject the Dirac hypothesis with $99.8\,\%$ CL (2.9σ)





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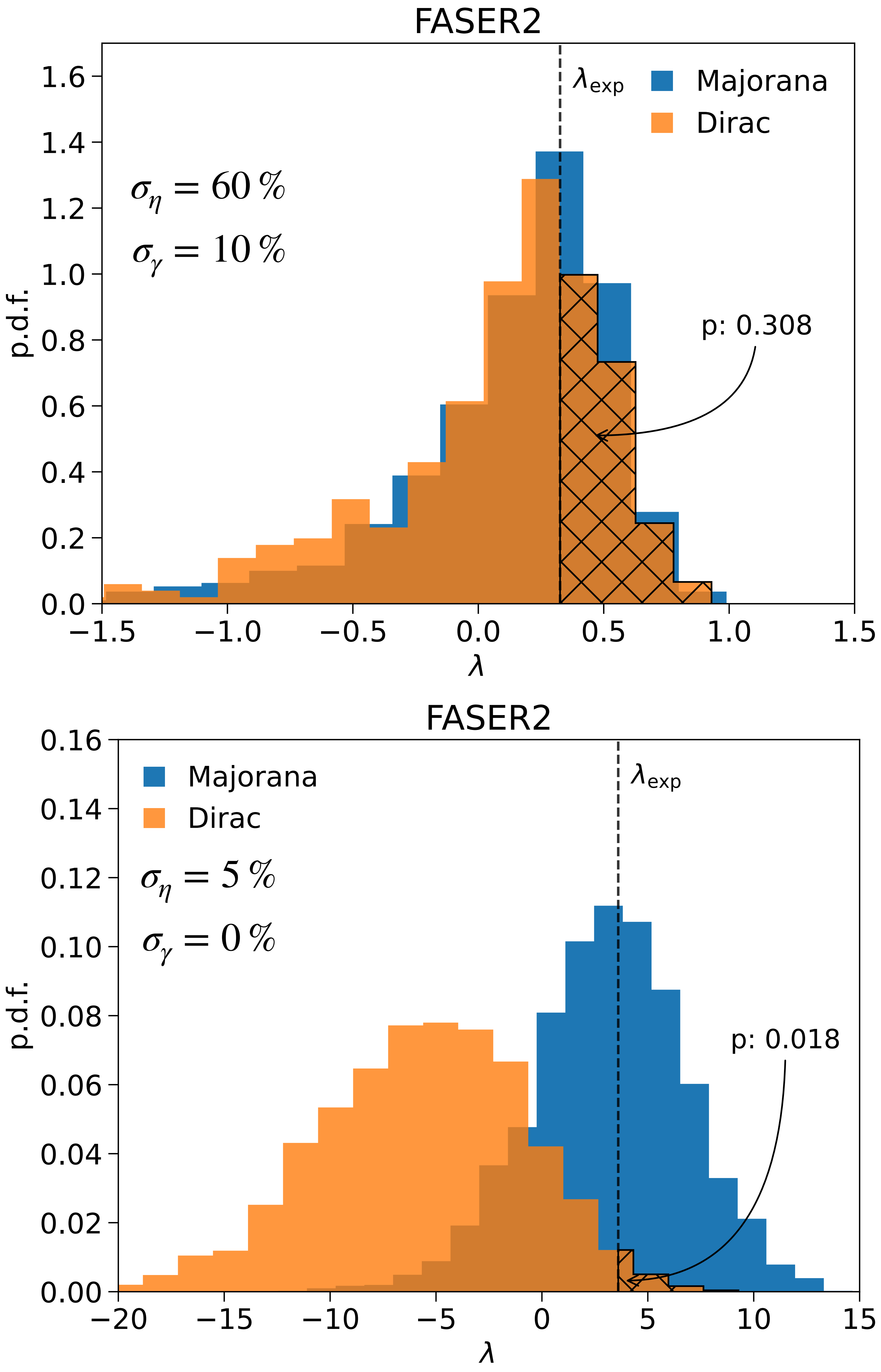
Discrimination via Kinematics

Model 2 (Low Statistics)

Profiled Likelihood Ratio Test

$$\lambda(\vec{d}\,) := -2\ln\frac{L_D(\vec{d}\,|\,\hat{m}_N^D, \hat{U}_\mu^D;\ \hat{\eta}^D, \hat{\vec{\gamma}}^D)}{L_M(\vec{d}\,|\,\hat{m}_N^M, \hat{U}_\mu^M;\ \hat{\eta}^M, \hat{\vec{\gamma}}^M)}$$



- However, with lower event rates, the procedure is **statistics limited**
- Assuming flux uncertainties are improved, we find that FASER2 can only expect to reject the Dirac hypothesis with $98.2\,\%$ CL (2.1σ)
- Therefore, in order to obtain comparable results to the high-statistics scenario, additional information is necessary.

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FASER2 as a Trigger for ATLAS

Model 2

Profiled Likelihood Ratio Test

$$\lambda(\vec{d}\,) := -\,2\ln\frac{L_D(\vec{d}\,|\,\hat{m}_N^D, \hat{U}_\mu^D;\, \hat{\eta}^D, \hat{\vec{\gamma}}^D)}{L_M(\vec{d}\,|\,\hat{m}_N^M, \hat{U}_\mu^M;\, \hat{\eta}^M, \hat{\vec{\gamma}}^M)}$$

- Assuming **improved flux uncertainties**, we find that FASER2 can expect to reject the Dirac hypothesis with $99.7\,\%$ CL (2.7σ), **despite having significantly lower statistics!**
- This illustrates that the main LHC detectors and proposed auxiliary detectors can have a synergistic relationship on an event-by-event basis.